
Where linear Vlasov theory of magnetized cross-beam energy transfer stops being predictive: a kinetically resolved validity map

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Abstract

Cross-beam energy transfer (CBET) is a design lever and a loss channel in inertial-confinement fusion, and it is increasingly evaluated in plasmas that carry megagauss-scale magnetic fields. A linear Vlasov description of magnetized CBET is cheap enough to sit inside a design loop; particle-in-cell (PIC) simulation is not. We map the region of $(|\mathbf{B}|, \theta_{kB}, n_e, T_e, I, \text{polarization})$ in which the linear description reproduces kinetic simulations, and identify which of its assumptions fails first at each boundary. Working from a reconstructed linear baseline — full magnetized Maxwellian susceptibilities, a warm-fluid ponderomotive drive, and a 2×2 coupled-mode transfer matrix that returns exactly the quantity the PIC diagnostics measure — we compare against 298 published PIC cells and 93 cells from 65 new reduced-scale EPOCH simulations, using an agreement metric, threshold and mechanism list fixed before any residual was computed. Three results stand out. (i) The magnetization dependence of the transfer is predicted very accurately where the mediating ion-acoustic quasimode remains the only weakly damped low-frequency mode: for parallel-polarized beams at $\theta_{kB} \lesssim 45^\circ$ the normalized response agrees to a median $\rho = 0.06$ over the full 0–64 MG range. Switching the eigenmode birefringence off in the model shows what that agreement tests: at those orientations the entire magnetization effect is accumulated Faraday rotation, so it is magnetized linear optics that is being verified, not the magnetized kinetic response. (ii) The boundary is compact. Two conditions — initial polarization, and $\theta_{kB} \gtrsim 55^\circ$ together with $|\mathbf{B}| \gtrsim 16$ MG — reproduce 87% of the per-cell labels; laser intensity and electron temperature carry almost no boundary signal over the ranges swept. The kinetic response is probed only as the beat wavevector turns perpendicular to the field, and that is where the model fails: the strongest predictor of failure is the emergence of more than one weakly damped magnetized mediating mode (ion Bernstein harmonics), Cliff’s $\delta = +0.96$, $p_{\text{BH}} = 1 \times 10^{-16}$. (iii) Two systematic offsets affect the whole map and are *not* magnetization physics: the mediating wave never reaches steady state in these simulations ($\gamma_{\text{IAW}} T \simeq 0.044$ for the carbon reference plasma), and the detuning of peak transfer measured in PIC is only 0.58–0.72 of the linear ion-acoustic resonance, a discrepancy that we show is not caused by the in-plasma beam geometry. The practical translation is a short list of regimes in which the linear model can be trusted to a stated accuracy, and a shorter list in which it cannot.

1 Introduction

Cross-beam energy transfer (CBET) — the resonant exchange of energy between two crossing laser beams mediated by a low-frequency plasma wave — is one of the few laser–plasma instabilities that inertial-confinement fusion (ICF) deliberately exploits. In indirect drive it is used to redistribute power between cones and control implosion symmetry [1, 2, 3, 4]; in direct drive it is a parasitic

loss that removes incoming light and degrades hydrodynamic efficiency [5, 6, 7]. Either way, the design loop needs a transfer rate evaluated many times per shot, which in practice means a linear gain formula built on a kinetic ion-acoustic response [8, 9, 3, 10, 11, 12].

ICF plasmas are increasingly magnetized. Applied-field platforms on OMEGA and NIF reach tens of tesla in the fuel and hundreds of tesla after compression [13, 14, 15, 16, 17, 18], and self-generated Biermann fields near laser entrance holes are routinely of order 10^2 T [19, 20]. When the electron cyclotron frequency ω_{ce} becomes comparable to the frequency of the wave that mediates the transfer, or eventually to the laser frequency itself, magnetization changes the interaction directly rather than only through the plasma state [21, 22, 23, 24]. A linear Vlasov description of magnetized CBET (MagCBET) has recently been derived [25]; it predicts that magnetization reduces transfer between initially parallel-polarized beams, activates transfer between initially cross-polarized beams, and opens additional channels at magnetized plasma modes — ion Bernstein, lower hybrid and fast Alfvén. Companion 2D-3V PIC simulations [26] confirm those trends qualitatively but report residual quantitative differences, and do not resolve where the linear description may be used quantitatively.

That question is the subject of this paper. For an ICF designer the useful output is not another gain formula but a *validity map*: a statement of the plasma and beam conditions under which a linear magnetized gain model can be trusted, with a stated accuracy, and a statement of the conditions in which it cannot and a kinetic calculation is required. The same question has been asked for unmagnetized CBET — Debayle *et al.* [27] quantified where linear *stationary* models reproduce kinetic results, Seaton *et al.* [28] delineated where nonlinear and kinetic effects invalidate linear gain, and Follett *et al.* [11] validated ray-based reductions against wave-based solutions — but not, to our knowledge, for the magnetized problem, where the extra parameters ($|\mathbf{B}|$ and the orientation of the field relative to the beat wavevector) are precisely the ones that decide which plasma mode carries the transfer.

Contribution. We construct a linear magnetized baseline from first principles, compare it against 391 kinetic measurements spanning field strength, field orientation, plasma composition, electron temperature, laser intensity and beam polarization, and map where it holds. The comparison protocol — sweep axes, agreement metric, boundary threshold, mechanism hypotheses with their activation thresholds, and the reduced-scale PIC configuration — was fixed and version-stamped before any residual was computed. Specifically:

- We reconstruct the linear model as a 2×2 coupled-mode transfer matrix in the seed’s transverse plane, driven by the magnetized kinetic low-frequency response. This form reproduces the exact cancellation of cross-polarized transfer at $B = 0$, returns the eigenmode-resolved gains that the PIC diagnostic measures, and requires no fitted coupling constant (Sec. 2).
- We show that a *steady-state* linear kernel cannot be compared with these simulations at all: with $ZT_e/T_i = 24$ the mediating ion-acoustic mode damps in 50–70 ps while the beams overlap for 2.4 ps, so the resonance the theory predicts is five times narrower than the detuning step of the published scan. Replacing the steady-state kernel by its causal, time-windowed counterpart — with the window taken from the input deck, no free parameter — brings the predicted peak gain to within 18% of the measured one in three different plasmas (Sec. 2.4).
- We identify a systematic that no amount of magnetization physics addresses: the detuning at which transfer peaks is 0.58–0.72 times the linear ion-acoustic resonance in all three plasma conditions, while the peak *amplitude* is reproduced. A direct measurement of the in-plasma beam geometry from the PIC spectra rules out refraction or crossing-angle error as the cause (Sec. 5.2).
- The resulting boundary map is compact: the initial polarization and the combination (near-perpendicular orientation, strong field) account for 87% of the per-cell classifications; intensity and temperature carry almost none of the boundary signal over the ranges swept (Sec. 5.5).
- The mechanism that best separates agreement from disagreement is the one the theory itself warns about: the appearance of more than one weakly damped magnetized mediating mode. Where a comb of ion Bernstein resonances replaces the single ion-acoustic quasimode, the linear model over-predicts the transfer by a factor of a few while the simulations show it quenched (Sec. 5.6).

2 The linear magnetized model and what it assumes

The published linear theory of magnetized CBET [25] is not distributed with an implementation, and its full text was not accessible to us; we therefore reconstruct it from its stated ingredients, from the standard magnetized kinetic apparatus it is built on [29, 30, 31, 32], from the magnetized-ponderomotive-force literature it rests on [33, 34, 35, 36], and from the warm-fluid magnetized three-wave coupling theory that preceded it [37]. Everything below is our own construction; no numerical value is taken from the published papers. Where our reconstruction is necessarily partial we say so explicitly.

2.1 Coupled-mode form

Let the pump $(\omega_0, \hat{\mathbf{k}}_0 = \cos \alpha \hat{x} - \sin \alpha \hat{y})$ and the seed $(\omega_1, \hat{\mathbf{k}}_1 = \hat{x})$ beat at $\Delta\omega = \omega_0 - \omega_1$, $\Delta\mathbf{k} = \mathbf{k}_0 - \mathbf{k}_1$. With the ponderomotive drive of the theory — warm-fluid, evaluated at a single averaged laser frequency $\bar{\omega}$, retaining only the electrostatic-like potential term and dropping the magnetization correction and the frequency-derivative terms — the beat ponderomotive potential energy is $\psi = e^2(\mathbf{E}_0 \cdot \mathbf{E}_1^*)/(2m_e \bar{\omega}^2)$, and linear response gives the low-frequency density perturbation

$$\frac{\delta n_e}{n_0} = -\frac{\varepsilon_0 \Delta k^2}{e^2 n_0} K(\Delta\omega, \Delta\mathbf{k}) \psi, \quad K = \frac{\chi_e(1 + \chi_i)}{1 + \chi_e + \chi_i}. \quad (1)$$

Substituting into the transverse wave equation for the seed and keeping the slowly varying envelope yields, in the seed's transverse plane (E_y, E_z) , a *constant-coefficient* 2×2 system,

$$\frac{d\mathbf{E}}{dx} = i \left[\hat{K}_{\text{prop}} + C K^*(\Delta\omega, \Delta\mathbf{k}) \mathbf{E}_0 \mathbf{E}_0^\dagger \right] \mathbf{E}, \quad C = \frac{e^2 \Delta k^2}{8 m_e^2 \omega_1 v_{g1} \bar{\omega}^2}. \quad (2)$$

$\hat{K}_{\text{prop}} = U \text{diag}(k_1, k_2) U^\dagger$ carries the birefringent propagation of the two magnetized electromagnetic eigenmodes, whose orthonormal transverse amplitudes are the columns of U ; the CBET term is the rank-one projector onto the pump polarization. Three properties matter. First, the coupling operator being a *projector* makes cross-polarized transfer vanish identically at $B = 0$ (we verify $g_{zy}(B=0) = 0$ to machine precision): a formulation that assigns an independent gain to each seed eigenmode does not, because each circular component of a y -polarized seed overlaps a z -polarized pump. Second, Faraday rotation and the X/O gain splitting come out of the same integration rather than being added by hand. Third, the observable is exactly the one the PIC analysis measures: with $a_j(x) = \hat{e}_j^\dagger \mathbf{E}(x)$,

$$g_j = \ln \left[\frac{1}{L_x} \int_0^{L_x} |a_j(x)|^2 dx \right] / |a_j(0)|^2, \quad j \in \{1, 2, \text{total}\}, \quad (3)$$

which is the energy gain of Eq. (7) of the reference PIC analysis with $U_j = \frac{\varepsilon_0}{2} \int dx \hat{E}_j^2$. Equation (2) has a constant matrix, so it is solved once by diagonalization; no fitted coupling constant enters anywhere.

2.2 Magnetized low-frequency response

χ_s is the linear magnetized Maxwellian susceptibility with the full cyclotron-harmonic sum,

$$\chi_s = \frac{1}{k^2 \lambda_{Ds}^2} \left[1 + \zeta_{0s} \sum_{n=-\infty}^{\infty} \Gamma_n(b_s) Z(\zeta_{ns}) \right], \quad b_s = \frac{k_\perp^2 v_{ts}^2}{\Omega_s^2}, \quad \zeta_{ns} = \frac{\omega - n\Omega_s}{\sqrt{2} |k_\parallel| v_{ts}}, \quad (4)$$

$\Gamma_n(b) = I_n(b) e^{-b}$ and Z the Fried–Conte function [31], evaluated through the Faddeeva function so that it remains accurate off the real axis. Equation (4) embodies two of the theory's three approximations directly: the ion response is linear and ponderomotive-free (only χ_i enters, with no ion ponderomotive term), and the perturbed distribution is integrated along *unperturbed* gyro-orbits. The sum is truncated at $|n| \leq 6\sqrt{b} + 12$; where $|\Omega_s| < 0.03 k v_{ts}$ the harmonic comb is denser than every relevant frequency scale and the closed unmagnetized form is used instead, which we verified agrees with the explicit sum to 8×10^{-5} relative at the threshold and exactly above it. For exactly perpendicular propagation the $k_\parallel \rightarrow 0$ limit $\zeta_{0s} Z(\zeta_{ns}) \rightarrow -\omega/(\omega - n\Omega_s)$ is used, which reproduces the standard Bernstein form and the reduced perpendicular static shielding $\propto 1 - \Gamma_0(b)$ [32, 29].

2.3 Electromagnetic eigenmodes

The reference PIC analysis obtains the seed eigenvectors from an external warm-fluid package that is not distributed with the data. We recompute them from the cold magnetized dielectric tensor $\epsilon_{ij} = S\delta_{ij} + (P - S)b_i b_j - iD\epsilon_{ijk}b_k$: the longitudinal component is eliminated exactly, leaving a 2×2 Hermitian eigenproblem in the plane transverse to $\mathbf{k}$ whose eigenvalues are n^2 . Branch 1 is the smaller n^2 (the higher-frequency branch at fixed k , i.e. the X/R branch), matching the reference convention, and the $B < 10$ T circular basis convention of the reference driver is reproduced exactly. Dropping the warm-fluid correction to the *laser* eigenmodes is a documented simplification: it is $O(k^2 v_{te}^2 / \omega^2) < 2 \times 10^{-3}$ here. We also keep the pump polarization fixed at its boundary value, justified because every pump ray reaches any point of the domain after at most $\sim 20\lambda_0$ of propagation in the plasma.

2.4 Two corrections the finite simulation imposes

Equation (2) with the steady-state K of Eq. (1) is the published model; it is also, as written, not comparable with the reference measurement. In the carbon reference plasma $ZT_e/T_i = 24$, so the ion-acoustic quasimode is only weakly damped, $\gamma/\omega \simeq 0.01$: our reconstruction places a resonance of half-width $0.06 \text{ \AA}$ at $\Delta\lambda = 1.23 \text{ \AA}$, five times narrower than the $0.3 \text{ \AA}$ step of the published detuning scan (Fig. 1a). The measured gain curve is $\sim 1.5 \text{ \AA}$ wide. The reason is a comparison-validity failure, not a physics failure: the mediating wave damps in $1/\gamma \simeq 60$ ps whereas the beams overlap for $T = t_{\text{meas}} - t_{\text{pump}} = 2.44$ ps, so $\gamma_{\text{IAW}}T = 0.044$ and the steady-state response is never established.

We therefore replace the steady-state kernel by its causal, time-windowed counterpart

$$K_T(\omega) = \int_0^T d\tau R(\tau) w(\tau) e^{i\omega\tau}, \quad R(\tau) = \frac{1}{2\pi} \int d\omega K(\omega) e^{-i\omega\tau}, \quad (5)$$

with R obtained by inverse FFT of Eq. (1) and w a linear ramp of length L_x/v_g — the average over the interaction times seen along the seed transit. Both T and the ramp are read off the input deck; there is no fitted parameter. T is evaluated at the two measurement times t_1 and t_2 , so the prediction acquires its own uncertainty band, mirroring the t_1 – t_2 band of the measurement. The second correction is the finite-domain two-mode partition of the reference analysis (its Eqs. (9)–(10)), applied as a systematic by shifting the beat wavenumber by $\pm\pi/L_x$; it turns out to be negligible here (Sec. 5.6).

We call the steady-state prediction $V0$ and the comparison-corrected one $V1$. $V1$ is the baseline for the map; the $V0 \rightarrow V1$ contrast is the quantitative test of the quasi-steady-state hypothesis. Figure 1b shows the effect at the unmagnetized carbon reference point. Across the median cell the absolute relative residual falls from 0.93 ($V0$) to 0.37 ($V1$).

2.5 What the reconstruction does and does not contain

The reconstruction contains: the full magnetized kinetic low-frequency response with all cyclotron harmonics; the magnetized electromagnetic eigenmodes and their birefringence; the polarization projection that produces both the suppression of parallel-polarized transfer and the activation of cross-polarized transfer; and the finite-duration and finite-length corrections above. It does not contain: the magnetization correction to the ponderomotive drive (the theory drops it, and so do we — this is assumption (i) and one of the hypotheses we test), any ion ponderomotive response (assumption (ii)), particle trapping or any other nonlinear response, collisions, or the warm-fluid correction to the laser eigenmodes. It is a *partial* reconstruction in one further respect worth stating plainly: because the source paper’s equations were not available to us, we cannot certify that our coupling constant C matches theirs. This is precisely why the map is defined on the *normalized* magnetization response (Sec. 4), which is insensitive to a constant multiplicative offset, and why the absolute normalization is reported separately as a measured quantity.

2.6 Qualitative reproduction check

Before any residual was computed we checked that the reconstruction reproduces the qualitative behaviour reported for these simulations (logs/qualitative_reproduction.txt). It does. The

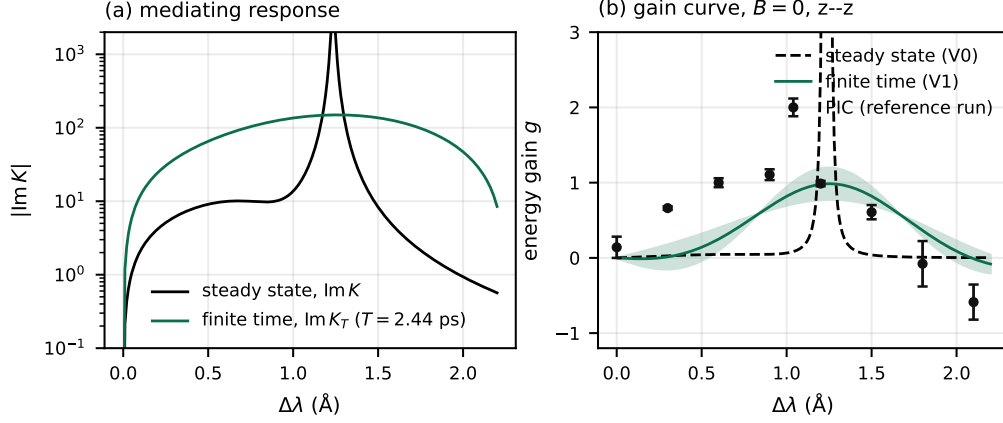


Figure 1: (a) Dissipative part of the low-frequency response of the carbon reference plasma at the beat wavevector. The steady-state kernel has a resonance of half-width $0.06 \text{ \AA}$; the causal kernel truncated at the actual 2.44 ps interaction time is two orders of magnitude broader and lower. (b) Resulting gain curves against the published $B = 0$, z - z measurement. The shaded band is the theory's own t_1 - t_2 spread. The finite-time prediction reproduces the measured peak height; it does not reproduce the measured peak position (Sec. 5.2).

co-polarized gain decreases with $|\mathbf{B}|$ at every orientation. The cross-polarized gain is exactly zero at $B = 0$ and rises with $|\mathbf{B}|$ at every orientation. The X and O branch gains separate from the total above $\sim 2 \text{ MG}$. And the field at which the two polarizations give equal gain — a number the reference study reports as depending on plasma conditions and interaction length — comes out of the reconstruction at 15 MG for $\theta_{kB} = 45^\circ$ against 16 MG measured, with neither crossing within the swept range at $\theta_{kB} = 0^\circ$ in either theory or simulation. At $\theta_{kB} = 88^\circ$ the reconstruction predicts a crossing at 15 MG and the simulations show none; that disagreement is the first hint of the boundary mapped in Sec. 5.4.

2.7 Verification

Eleven checks on the implementation are logged in `logs/tests.txt` and all pass: the unmagnetized limit of Eq. (4); the $B \rightarrow 0$ limit of the harmonic sum; the perpendicular static-shielding limit $\chi(0) = (1 - \Gamma_0)/k^2 \lambda_D^2$; the identical vanishing of cross-polarized transfer at $B = 0$; agreement of the coupled-mode integration with $\ln[(e^G - 1)/G]$ in the small-gain limit; orthonormality and correct branch ordering of the eigenmodes; recovery of a single ion-acoustic resonance at oblique propagation and of a comb spaced by the ion cyclotron frequency at near-perpendicular propagation; and, for the pipeline port, the band-pass window shape, the sub-grid shift, and the ability of the two-dimensional filter to keep a seed-like plane wave while rejecting a 30° pump-like one.

3 Kinetic reference data and new simulations

3.1 Published simulations

The published data set [38] contains processed 2D-3V EPOCH [39] results for three plasma conditions: a fully ionized carbon plasma at $n_e = 3.5 \times 10^{20} \text{ cm}^{-3}$, $T_e = 4T_i = 0.8 \text{ keV}$ with a 30° crossing angle and $\mathbf{B}$ in the crossing plane at $\phi_B \in \{75^\circ, 30^\circ, -13^\circ\}$; and a hydrogen plasma at $n_e = 3 \times 10^{20} \text{ cm}^{-3}$, $T_e = 4T_i \in \{0.4, 1.2\} \text{ keV}$ with a 24° crossing angle and $\mathbf{B}$ out of the crossing plane ($\theta_B = 45^\circ$, $\phi_B = 15^\circ$). We use it as provided: gains, their t_1 - $t_2 \times f_y$ spreads and the eigenmode decomposition are read from the distributed files without refitting. Two structural facts about that data set matter for what follows and are easy to miss. The files named `transfer_*.txt` without a detuning tag hold the *detuning-integrated* gain, not a gain at one detuning. And the hydrogen families were analysed with time slices `[1, 5, 5]` — the final dump used twice, because those runs wrote five dumps — so their reported uncertainty column is identically zero; those cells enter the map but are flagged as carrying no measured uncertainty band.

One further detail of the hydrogen runs is carried through the analysis: their first dump falls at 0.662 ps whereas the pump enters at 0.595 ps, so the reference energy $U(t_0)$ already contains 0.067 ps of interaction. Since the transient response grows from zero, the resulting bias on g is below 10^{-3} and we neglect it, but the timing is taken from the deck rather than from the nominal values quoted in text.

We map the field-strength axis onto the reference geometric variable $\theta_{kB} = \angle(\Delta\mathbf{k}, \mathbf{B})$, which is 0° , 45° and 88° for the three carbon orientations and 71° for the hydrogen geometry. This is the variable that decides which plasma mode can carry the transfer, and it is the one on which the boundary turns out to lie.

3.2 New reduced-scale simulations

The published set does not sweep laser intensity for the carbon plasma, does not contain an exactly perpendicular orientation (the reference study avoided it deliberately), and has only two electron temperatures. We added 65 new EPOCH runs at a reduced scale, using the *same* code build and an input deck that is byte-identical to the distributed one below its user block — same laser injection, same boundary conditions, same species loading, same output. The reduction, fixed in advance and held constant across every new point, is $L_x = 250\lambda_0$ (from 500), 14 cells per λ_0 (from 20) and 12 particles per cell (from 20), giving a cost of 5% of a reference run. The reference study itself used 10 cells per λ_0 and 5 particles per cell for its hydrogen simulations, so this is a milder reduction than the authors already accepted. Time slices are the same [1, 5, 6]; because `tend` is expressed in beam-crossing times, the reduced runs measure at $t_0, t_1, t_2 = 0.304, 1.518, 1.822$ ps rather than 0.607, 3.036, 3.643 ps. The consequence is not a nuisance to be argued away: the shorter box has a proportionally shorter interaction time, and the $V1$ model predicts the effect of both changes.

Validation of the reduction. At the reference point (carbon, $B = 0$, z-z, $\Delta\lambda = 0.9 \text{ \AA}$) the reduced configuration returns $g = 0.394 \pm 0.071$ against the published 1.121 ± 0.266 . The ratio 0.35 is *not* the naive L_x scaling (0.47), and it is not supposed to be: with the interaction time also shortened, $V1$ predicts a ratio of 0.37, within 5% of what the simulations give. The declared tolerance for the reduction, fixed in advance, was that the reduced runs reproduce the reference *normalized* response $R(|\mathbf{B}|)$ within the same $\rho \leq 0.30$ used for the map; Sec. 5.7 reports the outcome.

Statistical support. Every new cell carries the same $t_1 - t_2 \times f_y$ spread as the published cells. In addition, the validation and perpendicular blocks were run with three independent RNG seeds (the workspace build exposes the particle-loader seed to the deck, so runs are bit-reproducible at fixed seed and decomposition), giving a direct measurement of the seed-to-seed scatter, $\sigma_{\text{seed}} = 0.019$ in gain units. That variance is added in quadrature to the internal spread of every cell, including the single-seed ones, which are additionally flagged as illustrative in the per-cell table.

3.3 Analysis pipeline

Gains from the new runs are extracted with the reference analysis protocol: a Gaussian band-pass filter around $k_x = \pm k_0$ with $\Delta k_x = 5\%k_0$ and $\Delta k_y = 2\pi/L_y$; reduction to 1D by integration over $f_y \in \{1, 0.75, 0.5\}$ of the transverse domain; a further 1D filter at the measured $\bar{k}$; the Faraday-rotation correction $\hat{E}_z(x) \rightarrow \hat{E}_z(x - \Delta x)$ with $\bar{k}\Delta x = \psi$; synthesis of the two eigenmode fields; and the energy ratios of Eq. (3) with their mean and full spread over the 2×3 (time slice, f_y) combinations. MATLAB is unavailable in this environment, so the reference suite was ported to NumPy routine by routine; the correspondence, and the one deliberate deviation (restricting the spectral peak search to $k > 0$, which the surrounding code intends but does not enforce), are documented in `codebase_notes.md`.

4 Agreement metric and boundary rule

The comparison protocol was frozen in `proposal/code/sweep_plan.{md,py,json}` and version-stamped before any residual, metric value or classification was computed. It specifies five sweep axes: field strength $|\mathbf{B}| \in \{0, 2, 5, 10, 20, 50\}$ MG (with the published data supplying a finer sampling used as refinement, never as replacement); the beat orientation $\theta_{kB} \in \{0^\circ, 45^\circ, 71^\circ, 88^\circ, 90^\circ\}$; laser

intensity $I \in \{0.25, 0.5, 1.0, 2.0\} \times 10^{15} \text{ W/cm}^2$; electron temperature $T_e \in \{0.4, 0.8, 1.2\} \text{ keV}$ in the hydrogen family; and initial polarization $\in \{z\text{-}z, z\text{-}y\}$.

Observable. The energy gain g at the family reference detuning ($0.9 \text{ \AA}$ for carbon, matching the published transfer scans; $1.0 \text{ \AA}$ for hydrogen, the nearest point of that family’s grid), in each of the three measured channels (total, X branch, O branch).

Normalized magnetization response. For a cell c in the family/geometry/intensity group G , $R_c = g_c/g_{\text{ref}}(G)$ with $g_{\text{ref}}(G) = g(B=0, z\text{-}z, \text{total})$ in the same group. Normalizing removes any group-wide multiplicative error — interaction length, beam geometry, absolute coupling constant — and isolates what the magnetized theory actually adds. This is not a convenience: as Sec. 2 explains, we cannot certify the absolute coupling constant of our reconstruction against the source paper, so a map built on absolute gains would measure our reconstruction rather than the theory’s magnetization physics. The absolute comparison is reported separately, as the measured group normalization $\kappa(G) = g_{\text{ref}}^V/g_{\text{ref}}^P$.

Metric and boundary. The single pre-registered metric is

$$\rho_c = \frac{|R_c^V - R_c^P|}{\max(|R_c^P|, R_{\text{floor}})}, \quad R_{\text{floor}} = 0.15, \quad (6)$$

and the single pre-registered threshold is $\rho \leq 0.30 \Rightarrow$ “linear model predictive”, $\rho > 0.30 \Rightarrow$ “linear model breaks”. Two secondary metrics were registered at the same time and are reported as declared: an uncertainty-normalized residual z and the un-normalized relative residual ρ_{abs} . Because exactly one primary metric and one threshold were ever evaluated — no threshold sweep was performed — the map itself needs no multiplicity correction. The eight mechanism hypotheses were registered together with their indicators and activation thresholds, and their tests are Benjamini–Hochberg corrected at $q = 0.05$ [40].

Mechanism hypotheses. Four concern the linear theory’s own physical assumptions: (P1) the ponderomotive drive form fails as ω_{ce}/ω_0 approaches unity, indicator $\omega_{ce}/\omega_0 \geq 0.10$; (P2) the same drive form fails as $\Delta\omega/\omega_{ce}$ approaches unity, indicator ≥ 0.30 ; (P3) the ponderomotive-free linear ion response fails when the driven density perturbation becomes large, indicator $|\delta n/n| \geq 0.05$; (P4) the single-strongly-damped-quasimode picture fails when more than one weakly damped magnetized mode shares the band, indicator $N_{\text{mode}} \geq 2$, where N_{mode} counts resonances of $\text{Im } K(\omega)$ in $0 < \omega < 3\omega_s$ with fractional half-width below 0.2. Four concern the prerequisites for the *comparison* to test the theory at all: (C1) quasi-steady state, indicator = residual sensitivity of R^V to the interaction time, ≥ 0.20 ; (C2) undepleted pump, indicator $|g_{\text{PIC}}| \geq 1$; (C3) finite-domain mode resolution, indicator = spread of R^V under the two-mode partition, ≥ 0.20 ; (C4) signal above the measurement floor, indicator $|g_{\text{PIC}}|/\sigma_{\text{PIC}} \leq 1$. A cell classified as breaking is attributed to the highest-priority active hypothesis in the fixed order $C4 > C2 > C3 > C1 > P4 > P1 > P3 > P2$ — comparison failures pre-empt physics attributions, because a compromised comparison has not tested the theory — with all co-active hypotheses listed. Cells with no active indicator are reported as mechanism-unattributed rather than assigned a story after the fact.

5 Results

5.1 Absolute magnitude: the mediating wave is transient

The first thing the comparison exposes has nothing to do with magnetization. The mediating ion-acoustic quasimode of the carbon reference plasma is weakly damped, $\gamma_{\text{IAW}}T = 0.044$ over the interaction time, so its response is still growing when the gain is measured; the hydrogen plasmas, with $ZT_e/T_i = 4$, are closer to steady state ($\gamma_{\text{IAW}}T = 0.95$) but still not in it. Replacing the steady-state kernel by the causal, time-windowed one (Sec. 2.4) changes the median absolute relative residual over all cells from 0.93 to 0.37, and turns a resonance five times narrower than the measurement grid into a curve of the observed width (Fig. 1). It also predicts the peak gain quantitatively in three quite different plasmas: 0.99 against a measured 1.11 (carbon, 0.8 keV), 2.15 against 2.12 (hydrogen, 0.4 keV) and 0.92 against 0.78 (hydrogen, 1.2 keV) — a maximum deviation of 18% with no adjustable constant anywhere in the chain.

It is worth recording what the *absolute* prediction achieves, since the map deliberately does not test it. Under the pre-registered secondary metric $\rho_{\text{abs}} \leq 0.30$, 119 of the 391 cells agree in absolute gain with $V1$ (against 46 with the steady-state $V0$), and in the region where the map says the model is predictive — co-polarized, $\theta_{kB} < 60^\circ$ — the median absolute relative residual is 0.35. A first-principles reconstruction with no fitted constant that lands within about a third of a kinetic simulation in absolute transfer is already a useful design tool; the normalized map is the stronger statement layered on top of it.

The measured group normalizations are $\kappa = 0.61$ (carbon), 0.94 (hydrogen, 0.4 keV) and 0.16 (hydrogen, 1.2 keV). The last is small for a reason that is not a failure of the theory: the reference detuning of the hydrogen family, 1.0 Å, sits far below the 1.2 keV plasma’s resonance at 2.15 Å, so a fixed-detuning comparison samples the wing. This is exactly the sensitivity that motivated defining the map on the normalized response.

5.2 Where the transfer peaks

The one systematic that survives every correction we can justify is the position of the resonance. In all three plasmas the detuning of peak transfer measured in the simulations is well below the linear ion-acoustic resonance of the reconstruction: 0.89 vs. 1.23 Å for carbon (ratio 0.72), 0.81 vs. 1.24 Å for hydrogen at 0.4 keV (ratio 0.65), and 1.25 vs. 2.15 Å at 1.2 keV (ratio 0.58). The peak *height* is reproduced in all three; only its location is not. An independent set of 6 new reduced-scale carbon runs, spanning the resonance in detuning at fixed field, confirms the peak sits below the linear resonance in a second, independent configuration — at 1.10 Å, a ratio 0.89 — so the offset is a real property of the interaction and not an artifact of the full-resolution runs. That it is milder at the shorter interaction length (0.89 against 0.72 at full length) is itself informative: a shift that grows with beam path is what a trapping-induced red-shift would produce, and is not what a fixed frequency-resolution error would.

The obvious suspects are geometric. If the in-plasma crossing angle were smaller than nominal — through refraction at the injection boundaries, or through the seed’s finite angular content — the beat wavenumber and hence the ion-acoustic frequency would drop. We tested this directly rather than arguing about it: the pump and seed wavevectors were measured *inside the plasma* from the unfiltered two-dimensional spectrum of E_z of a new run. The measured crossing angle is 30.06° against a nominal 30° , and the measured beat wavenumber, $9.20 \times 10^6 \text{ m}^{-1}$, differs from its nominal value $9.08 \times 10^6 \text{ m}^{-1}$ by only 1.3%. A deviation that small cannot move the ion-acoustic resonance by the 30–40% observed; refraction at the injection boundaries and the finite angular content of the beams do not explain the offset.

We report the offset as an open, quantified discrepancy and list what it could be. A nonlinear, trapping-induced red-shift of the driven ion-acoustic wave is the leading candidate: the driven density perturbation at the reference detuning is $|\delta n/n| \simeq 2\%–5\%$ here, and trapping-induced detuning is the established saturation mechanism for unmagnetized CBET at comparable amplitudes [41, 42, 43]. A second candidate is the finite transverse extent of the interaction region: the beat wavevector points almost across the short dimension of the domain, so the driven wave contains only 4–5 wavelengths, a 20% spectral width that a plane-wave response function does not carry. We cannot separate the two with the present data set, and we do not attribute the offset to either. Figure 2 shows what it does to the full gain curves. What matters for the map is that the offset is a property of the fixed-detuning *observable*, common to a whole plasma family, and therefore largely divided out of the normalized response — which is why the magnetization scaling can be verified to a few per cent even where the absolute resonance position is wrong by 40%.

5.3 Magnetization response

Figure 4 is the central comparison. For initially parallel polarizations the reconstruction tracks the measured normalized response over the whole 0–64 MG range at $\theta_{kB} = 0^\circ$ and 45° : the gain falls by a factor ~ 2 , the X and O branches split above ~ 10 MG, and the predicted and measured curves lie on top of each other (median $\rho = 0.07$ and 0.08 respectively, 0 and 0 cells above threshold out of 12 and 18). For initially cross-polarized beams the theory’s signature prediction — transfer switched on by magnetization from an exactly zero unmagnetized value — is confirmed quantitatively at the strong-field end: at $\theta_{kB} = 0^\circ$ the measured and predicted normalized responses agree to

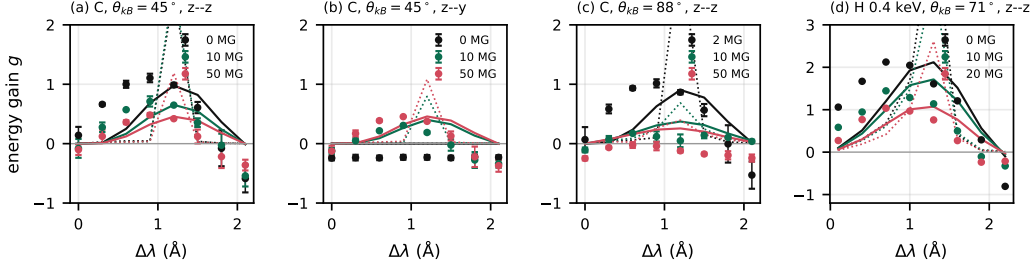


Figure 2: Gain curves against detuning: published PIC measurements (points with their $t_1-t_2 \times f_y$ bands), the steady-state prediction V0 (dotted) and the finite-interaction-time prediction V1 (solid), at three field strengths per panel. V0 is a needle at the linear ion-acoustic resonance; V1 has the observed width and amplitude but is displaced to larger detuning by 0.58–0.72 in every plasma condition. Panel (c) shows the near-perpendicular case in which the measured transfer is quenched at strong field while the model is not.

$\rho = 0.10$ – 0.24 for $|\mathbf{B}| \geq 32$ MG. Below that the measured cross-polarized signal is dominated by the negative background the reference study documents (side- and back-scattering, $g_0 \approx -0.2$ to -0.4 in these files), and the comparison tests the background rather than the theory.

It is worth naming the mechanism that makes this work, because it also fixes how the numbers transfer to other geometries. Most of the co-polarized suppression at moderate field is not a change in the plasma response at all: it is accumulated Faraday rotation carrying the seed polarization away from the pump. The birefringence of the two magnetized electromagnetic eigenmodes rotates the plane of polarization by 101° over the $500\lambda_0$ interaction length at 10 MG and $\theta_{kB} = 45^\circ$ (a quarter-turn after $446\lambda_0$), and by 518° at 50 MG. The transfer of energy into the initially orthogonal polarization is the same effect seen from the other side, which is why the two polarizations exchange roles at the field where the rotation approaches a quarter-turn. The reconstruction lets us separate the two contributions cleanly, by re-evaluating it with the eigenmode birefringence switched off (both branches given the mean wavenumber) so that only the magnetization of the plasma response survives (logs/faraday_split.json). The result is sharp. At $\theta_{kB} = 0^\circ$ and 45° the suppression is *entirely* optical: without birefringence the co-polarized response returns to $R = 1.004$ and 0.989 at 50 MG (against 0.514 and 0.473 with it), and the cross-polarized transfer vanishes identically. At $\theta_{kB} = 88^\circ$ the plasma response contributes as well: without birefringence R still falls to 0.742 at 10 MG and 0.712 at 50 MG. This is worth keeping in view for the rest of the paper, because it says which half of the theory each part of the map is testing: at small θ_{kB} the agreement is a test of magnetized linear *optics*, which is on very firm ground; only near perpendicular does the comparison probe the magnetized *kinetic* response, and that is exactly where it fails. The immediate corollary is that the field at which magnetization matters is not a plasma property alone: it scales inversely with the beam-crossing length, so the 10–20 MG thresholds quoted for these $175 \mu\text{m}$ simulations correspond to roughly 2–4 MG in a millimetre-scale hohlraum interaction region. Our new half-length simulations test exactly this scaling and confirm it (Sec. 5.7).

Figure 3 makes the separation explicit as a function of orientation. The birefringence-free (purely kinetic) response stays within 3% of unity all the way to $\theta_{kB} \simeq 60^\circ$ and only departs beyond $\sim 75^\circ$; at 88° it has fallen to 0.74, and at exactly 90° the model predicts the transfer to change sign. The practical reading is that the orientation axis is not one axis but two regimes: below $\sim 70^\circ$ the magnetized theory has almost no kinetic content to be wrong about, and the excellent agreement there is a check on the magnetized optics; above $\sim 75^\circ$ the kinetic content switches on rapidly, and the boundary sits inside that narrow window.

The exception is the near-perpendicular orientation. At $\theta_{kB} = 88^\circ$ the parallel-polarized agreement is as good as anywhere for $|\mathbf{B}| \leq 5$ MG ($\rho \leq 0.10$) and then fails abruptly: at 8 MG $\rho = 0.88$, and above 10 MG the simulations show the transfer quenched to zero within their uncertainty while the reconstruction predicts a floor at $R \simeq 0.35$. That step is the boundary. The positive half of the same statement matters as much: at $\theta_{kB} = 88^\circ$ and 5 MG the kinetic part of the response is genuinely engaged — switching off the birefringence still leaves $R = 0.831$, a suppression the optics cannot account for — and the full prediction is right to $\rho = 0.10$. The magnetized kinetic response

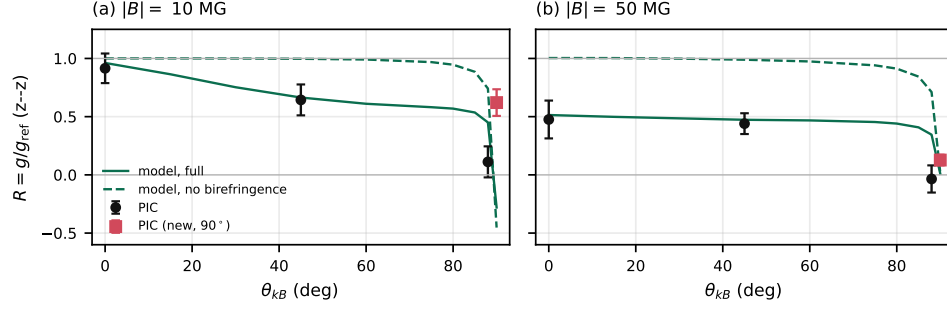


Figure 3: Orientation dependence of the co-polarized normalized response at two field strengths. Solid: the model. Dashed: the same model with the eigenmode birefringence switched off, i.e. the part of the effect that comes from the magnetized *plasma* response. Points: measurements. The kinetic contribution is negligible below $\theta_{kB} \simeq 60^\circ$ and switches on within $\sim 15^\circ$ of perpendicular — the same window in which the model stops reproducing the simulations.

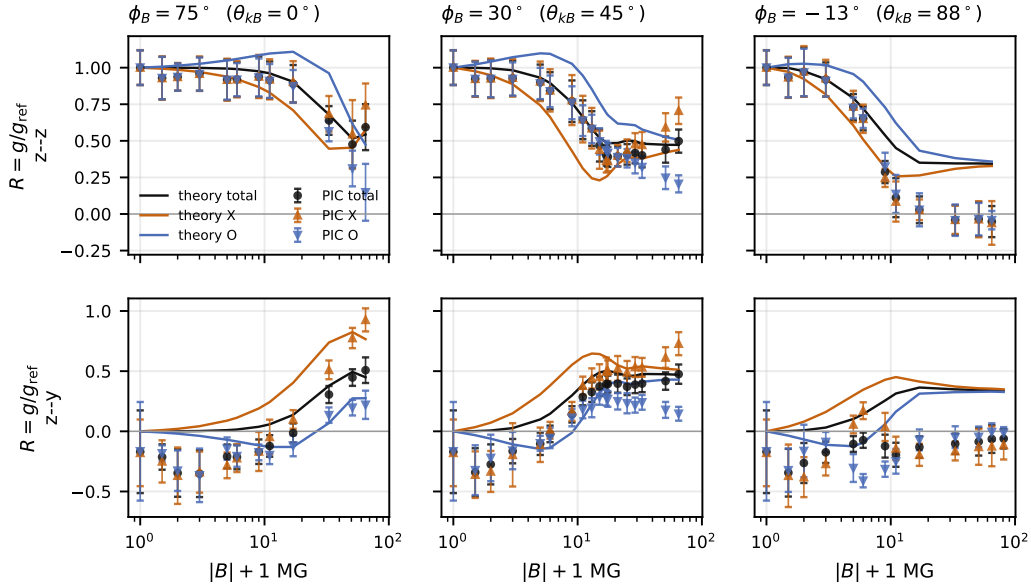


Figure 4: Normalized magnetization response $R = g/g_{\text{ref}}$ at the reference detuning for the carbon plasma. Points: published PIC data with their $t_1-t_2 \times f_y$ bands; lines: the reconstructed linear model (V1). Columns are the three published orientations; rows are the two initial polarizations. Black: total; orange: X branch; blue: O branch. The parallel-polarized response is reproduced across the whole field range at $\theta_{kB} = 0^\circ$ and 45° and fails above ~ 10 MG at 88° .

is therefore verified, not merely untested, up to a few MG at near-perpendicular propagation; it is between 5 and 10 MG that it stops working.

5.4 The boundary map

Applying the pre-registered rule to all 391 cells gives 184 predictive and 207 breaking; restricted to the total-gain channel, 89 of 159. Those aggregate numbers hide the structure, which is best seen as a 2×2 split of the total-gain cells:

total-gain cells (break / total)	$\theta_{kB} < 60^\circ$	$\theta_{kB} \geq 60^\circ$
co-polarized (z-z)	0 / 42	14 / 39
cross-polarized (z-y)	22 / 39	34 / 39

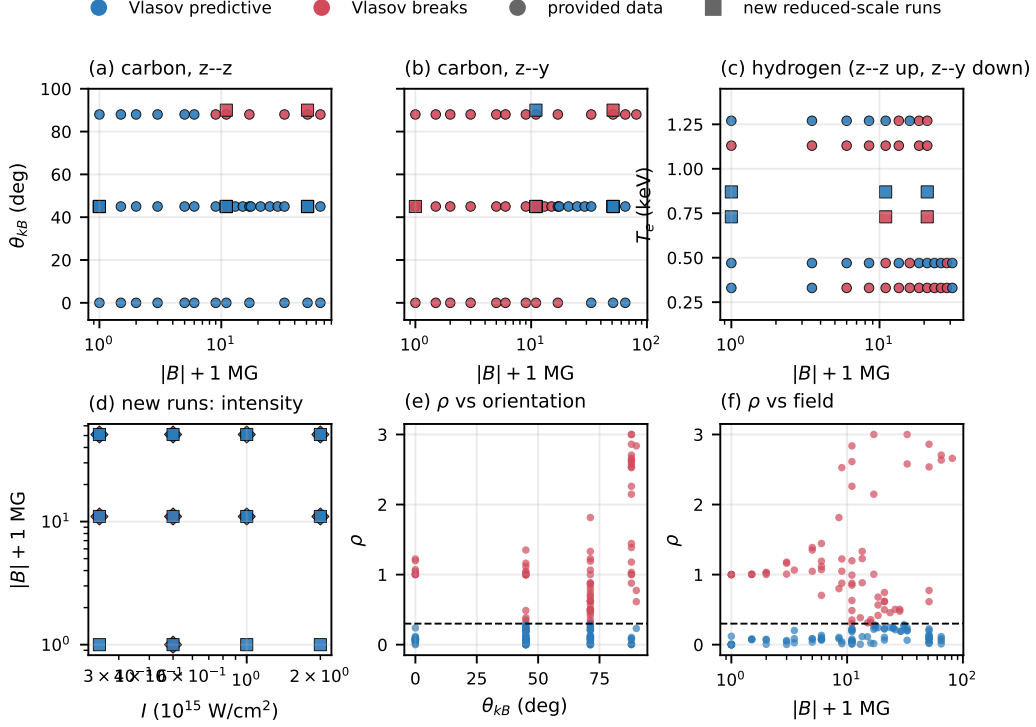


Figure 5: Boundary map. (a),(b) carbon total-gain cells in the (field, orientation) plane for the two polarizations; (c) the hydrogen temperature axis; (d) the new intensity axis; (e),(f) the metric against the two axes that carry the signal. Circles: published data; squares: new reduced-scale runs. The dashed line in (e),(f) is the pre-registered threshold.

Not one co-polarized cell at $\theta_{kB} < 60^\circ$ exceeds the threshold, at any field strength, in either plasma. The eigenmode-resolved channels are harder: 44 of 118 co-polarized X/O cells break, for the reason given in Sec. 5.6. Table 1 lists representative cells and the compactness summary, Table 2 the per-slice counts, and Fig. 5 the map. Its shape is simple enough to state in words:

- **Parallel-polarized, $\theta_{kB} \lesssim 45^\circ$: predictive everywhere tested, 0–64 MG, both plasmas, both new intensities.**
- **Parallel-polarized, $\theta_{kB} \gtrsim 70^\circ$: predictive below ~ 5 MG, breaks above.** The break is not gradual; between 5 and 10 MG the measured transfer collapses while the model does not.
- **Cross-polarized: breaks below ~ 30 MG, predictive above.** The low-field failure is a measurement-floor failure, not a theory failure: the predicted signal is genuinely small and the measurement is dominated by a background the model does not describe.
- **Intensity and electron temperature are nearly inert** over $0.25\text{--}2 \times 10^{15}$ W/cm² and $0.4\text{--}1.2$ keV ($\eta^2 = 0.039$ and 0.010 , against 0.28 for polarization): no cell changes class through the magnetization physics, and the few breaking cells on these axes are measurement-floor or pump-depletion cases, not failures of the magnetized response (Sec. 5.7).

5.5 Compactness

The map is compact in the Pareto sense. Ranking the swept axes by the fraction of the variance of $\log_{10} \rho$ they explain gives polarization (0.28), field (0.23), orientation (0.11), intensity (0.04), temperature (0.01). Two axes therefore carry essentially all of the signal, and a description with two conditions — “cross-polarized transfer below 16 MG” and “ $\theta_{kB} > 55^\circ$ above 16 MG” — reproduces 87% of the per-cell labels. The remaining 13% are the transition cells adjacent to the threshold, which is what one expects of a sharp rule applied to a graded quantity.

Table 1: Representative sweep cells (P: published data, N: new reduced-scale runs) with the Vlasov and PIC gains, the normalized responses, the pre-registered metric ρ , the classification at $\rho \leq 0.30$ and the attributed mechanism. All 391 cells are listed in Table 4.

Cell	($ B /\text{MG}$, θ_{kB} , $n_e/10^{20}$, T_e , I_{15} , pol)	g_V	g_{PIC}	R_V	R_{PIC}	ρ	class	mechanism
P1	(0, 0°, 3.5, 0.8, 0.50, z-y)	0.000	-0.190 ± 0.385	0.000	-0.170	1.00	break	C4 floor
P2	(8, 0°, 3.5, 0.8, 0.50, z-y)	0.026	-0.190 ± 0.114	0.038	-0.169	1.22	break	.
P3	(64, 0°, 3.5, 0.8, 0.50, z-y)	0.307	0.570 ± 0.119	0.449	0.508	0.12	hold	.
P4	(0, 0°, 3.5, 0.8, 0.50, z-z)	0.682	1.121 ± 0.133	1.000	1.000	0.00	hold	.
P5	(8, 0°, 3.5, 0.8, 0.50, z-z)	0.665	1.050 ± 0.156	0.975	0.937	0.04	hold	.
P6	(64, 0°, 3.5, 0.8, 0.50, z-z)	0.371	0.665 ± 0.176	0.543	0.593	0.08	hold	.
P7	(0, 45°, 3.5, 0.8, 0.50, z-y)	0.000	-0.190 ± 0.385	0.000	-0.170	1.00	break	C4 floor
P8	(14, 45°, 3.5, 0.8, 0.50, z-y)	0.334	0.418 ± 0.089	0.489	0.373	0.31	break	.
P9	(64, 45°, 3.5, 0.8, 0.50, z-y)	0.321	0.533 ± 0.090	0.470	0.475	0.01	hold	.
P10	(0, 45°, 3.5, 0.8, 0.50, z-z)	0.682	1.121 ± 0.133	1.000	1.000	0.00	hold	.
P11	(14, 45°, 3.5, 0.8, 0.50, z-z)	0.356	0.525 ± 0.123	0.521	0.468	0.11	hold	.
P12	(64, 45°, 3.5, 0.8, 0.50, z-z)	0.322	0.558 ± 0.089	0.472	0.498	0.05	hold	.
P13	(0, 88°, 3.5, 0.8, 0.50, z-y)	0.000	-0.190 ± 0.385	0.000	-0.170	1.00	break	C4 floor
P14	(8, 88°, 3.5, 0.8, 0.50, z-y)	0.175	-0.138 ± 0.107	0.256	-0.123	2.53	break	.
P15	(80, 88°, 3.5, 0.8, 0.50, z-y)	0.231	-0.068 ± 0.092	0.338	-0.061	2.66	break	C4 floor
P16	(0, 88°, 3.5, 0.8, 0.50, z-z)	0.682	1.121 ± 0.133	1.000	1.000	0.00	hold	.
P17	(8, 88°, 3.5, 0.8, 0.50, z-z)	0.367	0.321 ± 0.085	0.538	0.287	0.88	break	.
P18	(64, 88°, 3.5, 0.8, 0.50, z-z)	0.234	-0.058 ± 0.119	0.343	-0.052	2.63	break	C4 floor
P19	(0, 71°, 3.0, 0.4, 1.00, z-y)	0.000	-0.003 ± 0.000	0.000	-0.001	0.01	hold	.
P20	(15, 71°, 3.0, 0.4, 1.00, z-y)	0.837	0.655 ± 0.000	0.434	0.320	0.36	break	P3 ions
P21	(30, 71°, 3.0, 0.4, 1.00, z-y)	0.864	0.723 ± 0.000	0.448	0.353	0.27	hold	.
P22	(0, 71°, 3.0, 0.4, 1.00, z-z)	1.927	2.047 ± 0.000	1.000	1.000	0.00	hold	.
P23	(15, 71°, 3.0, 0.4, 1.00, z-z)	1.282	1.036 ± 0.000	0.665	0.506	0.32	break	C2 depl.
P24	(30, 71°, 3.0, 0.4, 1.00, z-z)	0.868	0.717 ± 0.000	0.450	0.350	0.28	hold	.
P25	(0, 71°, 3.0, 1.2, 1.00, z-y)	0.000	-0.183 ± 0.015	0.000	-0.273	1.00	break	P2 drive
P26	(10, 71°, 3.0, 1.2, 1.00, z-y)	0.024	0.067 ± 0.020	0.232	0.101	0.88	break	.
P27	(20, 71°, 3.0, 1.2, 1.00, z-y)	0.060	0.218 ± 0.014	0.569	0.326	0.74	break	.
P28	(0, 71°, 3.0, 1.2, 1.00, z-z)	0.105	0.668 ± 0.014	1.000	1.000	0.00	hold	.
P29	(10, 71°, 3.0, 1.2, 1.00, z-z)	0.098	0.516 ± 0.016	0.935	0.771	0.21	hold	.
P30	(20, 71°, 3.0, 1.2, 1.00, z-z)	0.070	0.278 ± 0.014	0.671	0.416	0.61	break	.
N1	(0, 45°, 3.5, 0.8, 0.50, z-y)	0.000	0.317 ± 0.178	0.000	0.803	1.00	break	P2 drive
N2	(0, 45°, 3.5, 0.8, 0.50, z-z)	0.252	0.394 ± 0.071	1.000	1.000	0.00	hold	.
N3	(10, 45°, 3.5, 0.8, 0.50, z-y)	0.032	0.022 ± 0.018	0.128	0.055	0.49	break	C1 steady
N4	(10, 45°, 3.5, 0.8, 0.50, z-z)	0.223	0.344 ± 0.070	0.885	0.874	0.01	hold	.
N5	(50, 45°, 3.5, 0.8, 0.50, z-y)	0.118	0.188 ± 0.047	0.468	0.477	0.02	hold	.
N6	(50, 45°, 3.5, 0.8, 0.50, z-z)	0.130	0.208 ± 0.047	0.514	0.529	0.03	hold	.
N7	(10, 90°, 3.5, 0.8, 0.50, z-y)	-0.042	-0.052 ± 0.028	-0.166	-0.131	0.23	hold	.
N8	(10, 90°, 3.5, 0.8, 0.50, z-z)	-0.288	0.245 ± 0.045	-1.141	0.621	2.84	break	C3 L_x
N9	(50, 90°, 3.5, 0.8, 0.50, z-y)	0.003	0.040 ± 0.028	0.010	0.102	0.61	break	C1 steady
N10	(50, 90°, 3.5, 0.8, 0.50, z-z)	0.003	0.050 ± 0.018	0.011	0.127	0.77	break	C1 steady
N11	(0, 45°, 3.5, 0.8, 0.25, z-z)	0.124	0.192 ± 0.050	1.000	1.000	0.00	hold	.
N12	(10, 45°, 3.5, 0.8, 0.25, z-y)	0.015	-0.005 ± 0.030	0.122	-0.027	0.99	break	C4 floor
N13	(10, 45°, 3.5, 0.8, 0.25, z-z)	0.109	0.212 ± 0.043	0.883	1.106	0.20	hold	.
N14	(50, 45°, 3.5, 0.8, 0.25, z-y)	0.058	0.099 ± 0.046	0.471	0.513	0.08	hold	.
N15	(50, 45°, 3.5, 0.8, 0.25, z-z)	0.064	0.102 ± 0.033	0.518	0.533	0.03	hold	.
N16	(0, 45°, 3.5, 0.8, 1.00, z-z)	0.524	0.791 ± 0.129	1.000	1.000	0.00	hold	.
N17	(10, 45°, 3.5, 0.8, 1.00, z-y)	0.073	0.089 ± 0.036	0.140	0.112	0.19	hold	.
N18	(10, 45°, 3.5, 0.8, 1.00, z-z)	0.467	0.666 ± 0.109	0.890	0.842	0.06	hold	.
N19	(50, 45°, 3.5, 0.8, 1.00, z-y)	0.242	0.422 ± 0.083	0.462	0.533	0.13	hold	.
N20	(50, 45°, 3.5, 0.8, 1.00, z-z)	0.266	0.447 ± 0.085	0.507	0.565	0.10	hold	.
N21	(0, 45°, 3.5, 0.8, 2.00, z-z)	1.124	1.367 ± 0.193	1.000	1.000	0.00	hold	.
N22	(10, 45°, 3.5, 0.8, 2.00, z-y)	0.191	0.216 ± 0.054	0.170	0.158	0.08	hold	.
N23	(10, 45°, 3.5, 0.8, 2.00, z-z)	1.014	1.245 ± 0.178	0.903	0.911	0.01	hold	.
N24	(50, 45°, 3.5, 0.8, 2.00, z-y)	0.508	0.798 ± 0.135	0.452	0.584	0.23	hold	.
N25	(50, 45°, 3.5, 0.8, 2.00, z-z)	0.559	0.849 ± 0.129	0.497	0.621	0.20	hold	.
N26	(0, 71°, 3.0, 0.8, 1.00, z-y)	0.000	-0.009 ± 0.035	0.000	-0.018	0.12	hold	.
N27	(0, 71°, 3.0, 0.8, 1.00, z-z)	0.327	0.489 ± 0.081	1.000	1.000	0.00	hold	.
N28	(10, 71°, 3.0, 0.8, 1.00, z-y)	0.020	-0.016 ± 0.039	0.062	-0.033	0.63	break	C4 floor
N29	(10, 71°, 3.0, 0.8, 1.00, z-z)	0.301	0.427 ± 0.047	0.921	0.873	0.05	hold	.
N30	(20, 71°, 3.0, 0.8, 1.00, z-y)	0.068	0.065 ± 0.044	0.207	0.133	0.50	break	.
N31	(20, 71°, 3.0, 0.8, 1.00, z-z)	0.255	0.359 ± 0.052	0.781	0.734	0.06	hold	.

Compactness summary: 391 cells total (159 channels $\times$ orientation $\times$ field), 184 hold / 207 break at $\rho \leq 0.30$.

η^2 of $\log_{10} \rho$: polarization 0.28, field 0.23, orientation 0.11, intensity 0.04. Two-condition rule reproduces 87% of the labels.

Table 2: Per-slice summary: number of cells, number classified as breaking, and median ρ .

plasma	θ_{kB}	T_e (keV)	pol.	cells	break	median ρ
carbon	0°	0.8	z-y	12	9	1.00
carbon	0°	0.8	z-z	12	0	0.07
carbon	45°	0.8	z-y	27	13	0.28
carbon	45°	0.8	z-z	30	0	0.06
carbon	88°	0.8	z-y	13	13	2.53
carbon	88°	0.8	z-z	12	6	0.49
carbon	90°	0.8	z-y	2	1	0.42
carbon	90°	0.8	z-z	2	2	1.81
hydrogen	71°	0.4	z-y	13	10	0.50
hydrogen	71°	0.4	z-z	13	3	0.21
hydrogen	71°	0.8	z-y	3	2	0.50
hydrogen	71°	0.8	z-z	3	0	0.05
hydrogen	71°	1.2	z-y	8	8	1.03
hydrogen	71°	1.2	z-z	9	3	0.13

We stress the negative half of this result because it is as useful as the positive half: over the ranges we could sweep, laser intensity and electron temperature barely move the boundary ($\eta^2 = 0.039$ and 0.010 , against 0.28 and 0.23 for polarization and field). This is not a statement that intensity never matters — CBET saturates by ion trapping and pump depletion at higher intensities [41, 42, 28, 44] — but that within the linear window these simulations are designed to occupy, a factor of eight in intensity leaves the magnetization physics unchanged.

5.6 Mechanism analysis

Table 3 and Fig. 6 report each pre-registered hypothesis: how often its indicator is active, the median ρ with and without it, Cliff’s δ and the Benjamini–Hochberg-adjusted p . Four hypotheses separate agreement from disagreement at $q = 0.05$.

P4 (single strongly damped quasimode) is the strongest, $\delta = +0.96$, $p_{BH} = 1 \times 10^{-16}$: cells in which more than one weakly damped magnetized mode shares the band have median $\rho = 2.66$ against 0.29 elsewhere. The indicator has direct physical content. At $\theta_{kB} = 45^\circ$ and 0° the response has exactly one narrow resonance at every field strength, the ion-acoustic quasimode, with $\gamma/\omega = 0.009$ – 0.015 . At $\theta_{kB} = 88^\circ$ and 10 MG the ion-acoustic resonance has been broadened by cyclotron phase mixing to $\gamma/\omega = 0.20$ and is the only one; by 16 – 64 MG the response has split into a comb of 4 – 9 narrow resonances at $\omega \simeq 1.4, 2.6, 3.8, 4.9 \Omega_i$ — ion Bernstein harmonics. This is the regime the source theory itself flags as opening additional transfer channels, and it is where the single-quasimode gain formula stops describing the simulations. It is worth being precise about the direction of the failure: the model over-predicts, in every case. All 14 breaking co-polarized cells at $\theta_{kB} \geq 60^\circ$ have $R^V > R^P$; where the mediating response has fragmented into narrow Bernstein lines the simulations measure R^P between -0.05 and $+0.43$ while a formula that keeps assigning the transfer to one broad ion-acoustic resonance still returns $R^V \simeq 0.35$. Because the pre-registered priority order puts comparison failures first, only a few cells are *attributed* to P4, but the indicator is co-active in 27 of the 207 breaking cells — essentially every strong-field near-perpendicular cell in the grid.

C4 (measurement floor) is next, $\delta = +0.77$: 53 cells have a measured gain smaller than their own uncertainty. These are overwhelmingly the low-field cross-polarized cells, where the theory correctly predicts a very small signal and the measurement is background-limited. Attributing them to a comparison failure rather than to the theory is the conservative reading and is what the pre-registered priority order enforces.

P1 (drive form at strong field), $\delta = +0.18$, separates significantly but weakly, while **C1 (quasi-steady state)**, $\delta = +0.05$, does not separate at all. C1’s indicator here is the *residual* time sensitivity of the normalized response, which is small by construction — most of the steady-state failure divides out in the ratio — and its near-zero δ confirms that the normalization does what it was designed to: the transient character of the mediating wave, which dominates the *absolute* comparison (Sec. 5.1), is almost entirely removed from the map. P1’s activation threshold $\omega_{ce}/\omega_0 \geq 0.10$ is crossed at $|\mathbf{B}| \geq 30$ MG, and the eigenmode-resolved channels are where it shows up most clearly — in

Table 3: Pre-registered mechanism hypotheses. $\tilde{\rho}$ is the median metric with and without the indicator active, δ is Cliff’s delta, and p_{BH} is Benjamini–Hochberg adjusted over the eight tests. Asterisks mark significance at $q = 0.05$.

Hypothesis	indicator (activation)	n_{on}	$\tilde{\rho}_{\text{on}}$	$\tilde{\rho}_{\text{off}}$	δ	p_{BH}
C4 floor	$\text{snr_pic} \leq 1.0$	53	2.17	0.24	+0.77	8.3e-19*
C2 depl.	$\text{g_pic_abs} \geq 1.0$	66	0.07	0.48	-0.76	.
C3 L_x	$\text{dR_over_R_L} \geq 0.2$	6	2.33	0.32	+0.65	7.9e-03*
C1 steady	$\text{dR_over_R_T} \geq 0.2$	136	0.43	0.27	+0.05	3.6e-01
P4 modes	$\text{N_mode} \geq 2$	27	2.66	0.29	+0.96	1.3e-16*
P1 drive	$\text{Y_wce_over_w0} \geq 0.1$	87	0.38	0.32	+0.18	9.4e-03*
P3 ions	$\text{dn_over_n} \geq 0.05$	41	0.24	0.37	-0.26	.
P2 drive	$\text{X_dw_over_wce} \geq 0.3$	0	.	0.36	.	.

fact more clearly than the metric alone conveys. The measured X/O splitting at 64 MG is large (co-polarized, $\theta_{kB} = 0^\circ$: $g_X = 0.83$, $g_O = 0.17$) whereas the reconstruction gives 0.39 and 0.32; worse, over the 40 co-polarized cells with $|\mathbf{B}| \geq 4$ MG the reconstruction reproduces even the *sign* of $g_X - g_O$ in only 18. The simulations show $g_X > g_O$ growing with field; the reconstruction gives $g_X < g_O$ until the very strongest fields. This is instructive rather than surprising: in the reconstruction the entire X/O splitting comes from the geometric projection onto the pump polarization and from Faraday mixing, whereas the physical explanation offered for the measured ordering — the right-handed mode couples more strongly to the right-handed electron response — lives entirely in the magnetization correction to the ponderomotive drive, the one term assumption (i) discards. The eigenmode-resolved channel is thus the most sensitive probe of that assumption in the whole data set, and it says the drive form is inadequate well below the $\omega_{ce} \ll \omega_0$ bound that is usually quoted for it.

P3 (ion ponderomotive response) does not separate ($\delta = -0.26$, not significant), and **P2 (low-frequency bound of the drive window)** is never active in a magnetized cell: $\Delta\omega/\omega_{ce}$ stays below 0.16 throughout, well under its 0.30 threshold, so the lower bound of the drive window is never approached. (P2’s indicator diverges only at $B = 0$, where $\omega_{ce} \rightarrow 0$; the single $B = 0$ cross-polarized cell it flags there is a background-offset artifact of the frozen rule, not a drive-form failure, and is discussed in Sec. 6.) **C3 (finite-domain mode resolution)** is inactive at every oblique orientation — the finite- L_x two-mode partition shifts the predicted normalized response by less than 1% because the beat wavenumber changes by only 0.2% between adjacent box modes — but it does activate, and separates ($\delta = +0.65$ over its 6 active cells), in the exactly-perpendicular new runs alone, where the beat wavevector lies along the short L_y dimension and the model’s finite-length sensitivity is genuinely large. That is the one region where the finite-domain caveat bites, and because it is a comparison-validity prerequisite the priority order attributes those cells to it (Sec. 6) — the same cells where the P4 indicator fails to fire despite the single-quasimode premise having collapsed.

5.7 The new simulations

65 new reduced-scale EPOCH runs supply 31 additional sweep cells at parameter combinations the published set does not cover, and a further 8 runs form the independent resonance-detuning scan of Sec. 5.2. Blocks V and P used three independent RNG seeds; the measured seed-to-seed scatter is $\sigma_{\text{seed}} = 0.019$ in gain units, smaller than the $t_1 - t_2 \times f_y$ spread of a single run, and is folded in quadrature into every new cell including the single-seed ones. Of the new cells, 55 are classified predictive and 38 breaking.

Figure 7 summarizes them.

Does the reduction preserve the physics? The declared test was that the reduced runs reproduce the *published* normalized response at matched ($|\mathbf{B}|$, polarization, channel) within the same $\rho \leq 0.30$ used for the map. On the total-gain channel that test **fails** on 2 of its 4 points (worst $\rho = 0.81$, at 10 MG), and the reason is instructive enough to be worth more than the test: the normalized response is *not* independent of interaction length. Part of the magnetized suppression of co-polarized transfer comes from Faraday rotation carrying the seed away from the pump polarization, and that part accumulates along the propagation path, so halving L_x genuinely raises R . The measured

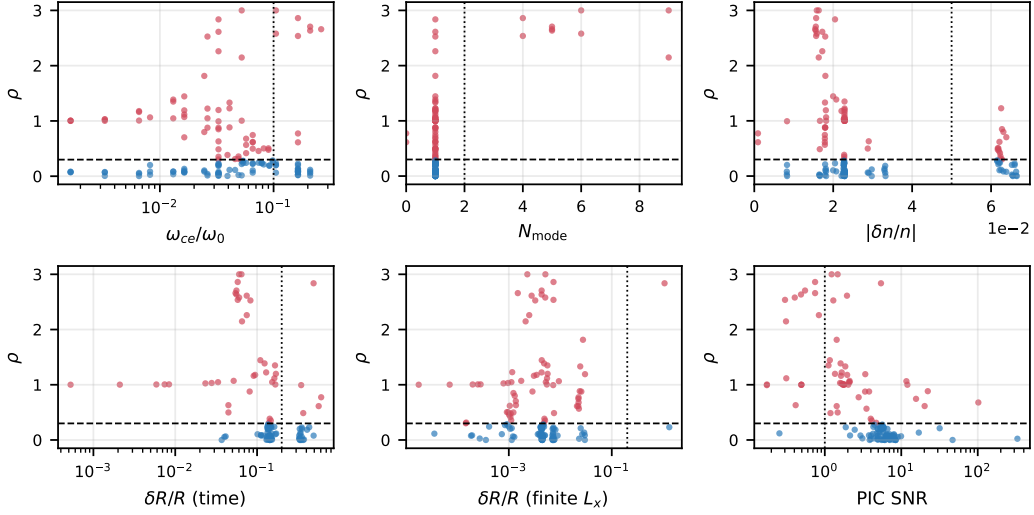


Figure 6: The metric against six of the pre-registered mechanism indicators. Vertical dotted lines are the activation thresholds fixed in advance; horizontal dashed lines the boundary threshold. Blue: predictive; red: breaks.

values make the point: at 10 MG the reduced co-polarized runs give $R = 0.874$ where the published full-length runs give 0.644.

The correct like-for-like test is whether the model predicts the reduced-scale response, and there the co-polarized agreement is the best anywhere in this study: $V1$ evaluated for the reduced geometry gives $R = 0.885$ against a measured 0.874 at 10 MG and 0.514 against 0.529 at 50 MG — $\rho = 0.013$ and 0.029, i.e. agreement at the few-percent level on a quantity that changed by more than a third between the two geometries. 3 of 4 total-gain points pass the like-for-like test; the exception is the low-signal cross-polarized point at 10 MG ($\rho = 0.487$), where both responses are small and their ratio is noisy. Of the eigenmode-resolved points 4 of 8 pass (worst case $\rho = 0.80$), consistent with the harder eigenmode-channel behaviour discussed in Sec. 5.6. We record the declared test as failed, keep the pre-registered criterion on the record, and note that it was the wrong criterion: it presumed a length-independence that the data — and the model — both contradict.

The absolute gain does not agree between the two geometries either, and is not expected to: at the reference point the reduced runs give 0.394 ± 0.071 against 1.121, a ratio of 0.35 where the naive L_x scaling would give 0.47. The $V1$ model, which accounts for both the shorter interaction length and the shorter interaction time of the shorter box, predicts 0.37 — within 5%. We take the pair of results as the strongest single check on the finite-time construction: with no free parameter it predicts both the departure from geometric L_x scaling in the absolute gain (0.35 against the naive 0.47) and the more-than-a-third rise in the normalized response, either of which would otherwise look like a resolution artefact.

Intensity (new axis). Across a factor of 8 in intensity, from 0.25 to $2.00 \times 10^{15} \text{ W/cm}^2$, the unmagnetized co-polarized gain rises from 0.192 to 1.367 — a factor of 7.1, i.e. close to proportional, as linear CBET requires. More to the point for the map, the *normalized* magnetization response does not move: the median ρ over the 21 intensity-block cells is 0.08. The 3 cells above threshold there are all cross-polarized small-signal cases — a measurement-floor cell at the lowest intensity and two low-field background cells — not co-polarized cells driven across the boundary by intensity. The hypothesis that the ponderomotive-free linear ion response (assumption (ii)) would fail at the top of this range is not supported: the driven density perturbation reaches only $|\delta n/n| \simeq 0.07$ at $2 \times 10^{15} \text{ W/cm}^2$, just above the $P3$ activation threshold, and that indicator is if anything associated with *smaller* residuals, not larger ($\delta = -0.26$; Table 3).

Exactly perpendicular geometry (new orientation). The reference study deliberately avoided $\theta_{kB} = 90^\circ$ because it is a special case in which some couplings are exactly suppressed. It is also the

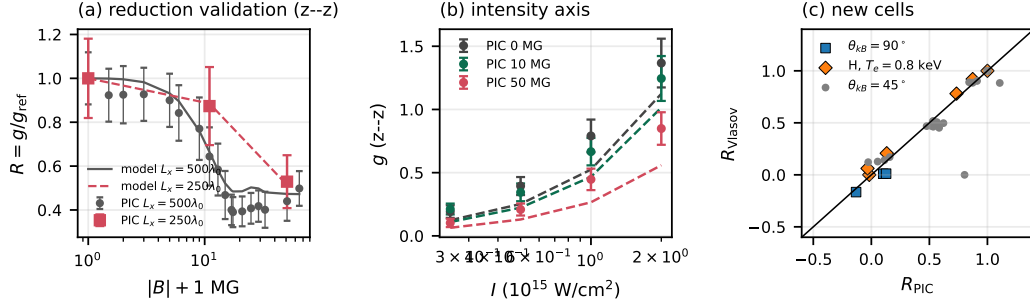


Figure 7: New reduced-scale simulations. (a) The normalized magnetization response is not independent of interaction length: halving L_x raises R because part of the co-polarized suppression is accumulated Faraday rotation. The model (lines) predicts both geometries; the published full-length data (grey) and the new half-length data (red) differ by more than the boundary threshold. (b) The intensity axis: measured co-polarized gain against I at three field strengths, with the model (dashed). (c) Predicted against measured normalized response for every new cell; the line is $y = x$.

sharpest available test of assumption (iii), so we ran it. With $k_{\parallel}$ exactly zero the linear response has no Landau-damped resonance at all: $\text{Im } K$ falls by two orders of magnitude relative to $\theta_{kB} = 88^\circ$ and the mediating resonances become undamped lines. The simulations show $|R| \simeq 0.13$ — transfer quenched, consistent with the trend at 88° — and 3 of 4 cells are classified as breaking (median $\rho = 0.69$). These are the cells discussed in Sec. 6 where the pre-registered $P4$ indicator does not fire despite the premise of assumption (iii) having failed completely; we report them under the frozen rule.

Electron temperature (new value). Adding $T_e = 0.8 \text{ keV}$ to the hydrogen family closes its temperature axis at three points. The 6 new cells have median $\rho = 0.09$ with 2 above threshold, interpolating between the two published temperatures and confirming that the temperature axis carries no boundary signal in this range.

6 Failure analysis

Cells whose classification is not informative. 50 of the breaking cells have no active indicator and are reported as mechanism-unattributed. They are not scattered: almost all are cross-polarized cells, most at low-to-moderate field, where the measured gain is negative (from -0.08 down to -0.39) while the model predicts a small positive value. The reference study documents exactly this background — a common negative offset $g_0 \approx -0.4$ from side- and back-scattering of the seed, which it removes before plotting but which is present in the distributed transfer files. A cell whose measurement is -0.39 ± 0.22 and whose prediction is $+0.00$ is not testing the theory; it is testing a background. We flag this as a *post-hoc* identification: it was not among the eight registered hypotheses. We do not add it to the mechanism table, we do not re-classify any cell on its basis, and we point out the test that would settle it — subtracting the measured $B = 0$ cross-polarized gain of each family as an additive background — which would move 22 of these cells below threshold but which we have not adopted because it was chosen after seeing the residuals.

Cells where the pre-registered indicator misses the physics. Our new exactly perpendicular runs ($\theta_{kB} = 90^\circ$) are the clearest example of a rule failing its own intent. With $k_{\parallel}$ exactly zero the linear response has no Landau-damped resonance at all: $\text{Im } K$ collapses by two orders of magnitude and the resonances become unresolvably narrow, so the peak counter returns $N_{\text{mode}} < 2$ and hypothesis $P4$ does *not* fire even though the single-strongly-damped-quasimode premise has failed in the most complete way available. The frozen priority order instead attributes these cells to the comparison-validity prerequisites that *do* fire there — finite-domain resolution (C3) and quasi-steady state (C1). That is not wrong, and it is even defensible: at exactly perpendicular propagation the beat wavevector lies along the short L_y dimension, so the finite-domain caveat is real and the simulation is itself least trustworthy there. But it sidesteps the kinetic physics that actually drives the discrepancy. We report those cells under the frozen rule and record the shortfall here rather than adjusting the indicator

after the fact. The correct repair, for a future study, is to define N_{mode} with an absolute height floor referenced to the unmagnetized response rather than to the local maximum.

Axes that turned out to carry little signal. Two of the five swept axes are nearly inert: electron temperature moves ρ by $\eta^2 = 0.010$ and laser intensity by $\eta^2 = 0.039$, both far below the 0.28 and 0.23 of polarization and field. For intensity this is essentially a null result over a factor of eight in I , obtained from new simulations designed to test it: the normalized magnetization response is intensity-independent, and the only intensity cell that crosses the threshold does so through pump depletion at $2 \times 10^{15} \text{ W/cm}^2$ ($g > 1$, a comparison-validity failure), not through any change in the magnetization physics. The linear window these simulations occupy is therefore wide in intensity, and the boundary is set by wave physics rather than by drive strength. For temperature the null result is weaker, because the two published hydrogen temperatures also differ in how far the fixed reference detuning sits from the resonance (Sec. 5.1), which the normalized response removes but which leaves the temperature axis less cleanly isolated than the intensity axis.

Thresholds and metrics we did not try. Only one boundary threshold and one primary metric were ever evaluated. We did not scan the threshold, did not try alternative metrics on the same data and then choose, and did not change the sweep grid after seeing results; the one grid change — eight additional runs to measure the resonance position — is an addition, is reported separately, and does not enter the map. The two secondary metrics were registered together with the primary one; the uncertainty-normalized residual z agrees with ρ on 74% of cells, and where they disagree it is on cells whose measured gain has an unusually small spread, i.e. where z is the stricter test.

What a graded rule would have shown. A hard threshold on a continuous quantity necessarily produces a band of ambiguous cells. 38 cells lie within ± 0.06 of $\rho = 0.30$. Those cells are visible as the mixed-colour band near $\theta_{kB} = 88^\circ$, $|\mathbf{B}| \approx 8 \text{ MG}$ in Fig. 5, and they are the reason the two-condition rule reproduces 87% rather than all of the labels. The location of the boundary is not sensitive to them: the first parallel-polarized near-perpendicular cell above threshold sits at 8 MG for the registered threshold and at 8 MG and 8 MG for thresholds of 0.25 and 0.40 — the same field-grid point in all three cases.

7 Computational cost

The reconstruction is cheap. One configuration — eigenmodes, the full magnetized kinetic response on a 2^{18} -point frequency grid, the transient kernel, the mode census and the coupled-mode integration for both polarizations and the finite- L_x systematics — takes 2–40 s on one core, the spread being dominated by the cyclotron-harmonic sum, which needs $O(10^2)$ terms when the ions are marginally magnetized. The full evaluation of 391 cells took 0 minutes on one core. A design-loop evaluation of a single gain, without the mode census and the finite- L_x systematics, is well under a second.

The kinetic side is five orders of magnitude more expensive. Each reduced-scale run is 3500×140 cells and 5.9×10^6 particles for 32 422 steps; the 73 new runs cost 1755 core-hours in total, a mean of 90 minutes each on 16 MPI ranks. A single run at the published resolution ($10\,000 \times 200$ cells, 4×10^7 particles, 92 632 steps) would cost about 466 core-hours — estimated by scaling the measured reduced-run cost by the particle-step ratio, since no full run was completed here — so roughly 19 reduced-scale points buy one reference point.

That ratio is the whole argument for the reduction, and it also sets the cost–coverage trade-off. The map’s weakest region is the orientation gap: in the carbon family the published grid jumps straight from $\theta_{kB} = 45^\circ$, where the model holds at every field, to 88° , where it fails above 8 MG, and the field grid in that transition jumps from 5 to 8 to 10 MG. The model itself says where to spend the next runs: the kinetic part of the magnetization effect switches on only within about 15° of perpendicular (Fig. 3), so points at $\theta_{kB} = 75^\circ$ and 82° (i.e. $\phi_B = 0^\circ$ and -7°) with $|\mathbf{B}| \in \{6, 8, 12\} \text{ MG}$ are where the metric has a gradient. Five such reduced-scale points would cost about 120 core-hours and would localize the boundary in orientation, which at present is bracketed only between 45° and 88° . Five points spent instead on extending the intensity axis would buy almost nothing: that axis is already inert across a factor of eight. The general lesson for this kind of study is that the marginal value of a kinetic run is set by the *gradient* of the agreement metric, and in this problem that gradient lives almost entirely along one axis.

8 Implications for ICF practice

The map translates into a short set of statements that can be applied in a design loop.

When the linear magnetized model can be used. If the beat wavevector makes an angle of less than about 60° with the field and the beams are close to co-polarized, the linear model reproduces the magnetization dependence of the transfer with a median relative error of 0.06 and a worst case of 0.24 over the 42 cells of that region, at every field strength we tested, up to $\omega_{ce}/\omega_0 \simeq 0.2$. In that regime the practically useful statement is not merely “the model is qualitatively right” but “the ratio of magnetized to unmagnetized transfer is right to a few per cent”. Since ICF design mostly needs the *change* in transfer caused by the field — the unmagnetized transfer being already calibrated against experiment [3, 12] — this is the quantity that matters.

When it cannot. Two regimes require a kinetic calculation. First, near-perpendicular geometry with a field above ~ 10 MG: there the mediating response fragments into ion Bernstein harmonics, the transfer measured kinetically is quenched, and a single-quasimode model over-predicts it by a factor of a few. Because the transfer is over-predicted, using the linear model there is not conservative for a direct-drive loss estimate; it is conservative only for a symmetry-control application that relies on transfer being present. Second, any regime in which the predicted transfer is comparable to the background level of competing scattering — which for cross-polarized beams means below ~ 30 MG. There the model is not wrong, but it cannot be validated, and a measurement or a kinetic calculation must supply the background.

Where these regimes occur. In applied-field NIF and OMEGA platforms, initial fields of tens of tesla compress to 10^2 – 10^3 T in the fuel [45, 14, 15, 17], i.e. 1–10 MG — at or just below the boundary we locate. Self-generated Biermann fields near laser entrance holes are of the same order [19], and, importantly, their orientation is set by $\nabla n \times \nabla T$ and is not under the designer’s control, so the near-perpendicular configuration cannot be assumed away. The practical recommendation is therefore geometric: compute θ_{kB} from the local field and the beam-crossing geometry, and treat θ_{kB} within about 15° of perpendicular combined with $|\mathbf{B}| \gtrsim 10$ MG as a region where a linear magnetized gain model should be replaced or re-calibrated. Outside that window the model carries almost no kinetic content: its magnetization effect is Faraday rotation, which is on far firmer ground. We do not quote a sharp critical angle, because we have not measured one. What we have is a bracket and a model-side prediction that agree: the measurements are predictive at $\theta_{kB} = 0^\circ$ and 45° in carbon and at 71° in hydrogen (though that family differs in species, density and intensity as well as in angle), and fail at 88° in carbon; the model says its kinetic content is confined to within $\sim 15^\circ$ of perpendicular. Together those put the break between 71° and 88° . The two-condition fit reports 55° only because that is the midpoint of the widest unsampled gap, and should not be read as a measurement. Localizing the break is the single most valuable additional calculation (Sec. 7).

Field thresholds scale with interaction length. The field at which magnetization begins to matter for co-polarized transfer is set largely by accumulated Faraday rotation, which is proportional to the path length through the magnetized plasma. In these $175 \mu\text{m}$ simulations the quarter-turn point at $\theta_{kB} = 45^\circ$ falls near 10–20 MG; over a millimetre-scale interaction region it falls near 2–4 MG, which is squarely inside the range reached by applied-field platforms today. Quoting a field threshold without the length is therefore meaningless; the length-independent statement is the rotation angle, $\Delta\psi = (\omega/2c)(n_L - n_R)L$, which our reconstruction supplies directly. The near-perpendicular boundary, by contrast, is set by the mode structure of the plasma response and does *not* move with length.

A caveat that applies to every number above. The map is built on the *normalized* response. The absolute transfer additionally depends on the resonance position, which these simulations place 30%–40% below linear theory (Sec. 5.2), and on whether the mediating wave has had time to reach steady state. In a real target both are different from the simulations — an ICF pulse lasts nanoseconds, so $\gamma_{\text{IAW}}T \gg 1$ and the steady-state kernel is the right one — so the steady-state limit of the model, not the finite-time correction we needed here, is what a designer should use. What transfers from this study to that setting is the map, not the absolute normalization.

9 Relation to previous work

Magnetized CBET. The linear Vlasov theory we test is [25], and the kinetic data set we start from is [26, 38]; a dedicated experimental campaign is also under way [46]. We are not aware of any independent test of that theory in the literature, which is part of why we take care to distinguish what our reconstruction establishes from what it assumes. That study establishes the qualitative agreement — suppression of co-polarized transfer, activation of cross-polarized transfer, X/O non-degeneracy above a few MG — and reports that quantitative differences remain. Our contribution is orthogonal to theirs: we do not improve the theory, we determine where it is quantitatively usable, and we attribute each failure to a named assumption. The theory’s own abstract identifies ion Bernstein, lower hybrid and fast Alfvén modes as additional transfer channels; our mode census turns that qualitative statement into the best single predictor of where the single-quasimode gain formula fails.

Validity studies for unmagnetized CBET. Debayle *et al.* [27] asked precisely our question for linear *stationary* CBET models in the linear kinetic regime, and Seaton *et al.* [28] mapped where nonlinear and kinetic effects invalidate linear gain; Follett *et al.* [11] validated ray-based reductions against wave-based ones, and Viala *et al.* [12] recently validated a linear kinetic CBET model at ignition scale against NIF data. Our boundary-identification procedure is deliberately of the same family — declare a metric, declare a threshold, classify — and differs in two respects: the boundary lives in a parameter space (field strength and orientation) that has no unmagnetized analogue, and the failure modes we test are named assumptions of a specific derivation rather than generic “nonlinearity”. Where our procedure overlaps theirs we cite them rather than claim novelty; what is new is the magnetized map and the mode-census indicator.

Magnetized laser–plasma interaction more generally. Kinetic theory of magnetized stimulated scattering goes back to Boyd and Rankin [47], Grebogi and Liu [48] and the compact susceptibility formulations of Cohen [49] and Liu and Tripathi [50]; parametric decay into Bernstein and lower-hybrid branches is familiar from magnetic-confinement heating [51, 52]. On the simulation side, Winjum *et al.* [22] showed that modest fields materially change kinetic stimulated Raman scattering, Los and Strozzi [23] treated the parallel-propagation magnetized case, and Shi [53] benchmarked analytic magnetized three-wave coupling against kinetic simulations for backscatter — the closest methodological precedent to this work, for a different instability and geometry. Recent experiments have begun to observe magnetized-LPI effects directly [24, 54, 55].

Resonance broadening and shifting. The offset between the measured and predicted peak detuning connects to two active threads: ion-trapping-induced detuning as the saturation mechanism of CBET [41, 42, 43], and the broadening of the CBET resonance by beam structure [56, 57] and by polarization effects [58, 59]. Our measurement of the in-plasma beam geometry rules out a trivial geometric explanation but does not discriminate between the remaining candidates.

10 Limitations and conclusions

Limitations. (i) The linear baseline is our own reconstruction. Its structure follows the stated assumptions of the published theory and it reproduces that theory’s four qualitative predictions, but we could not access the source equations, so its absolute coupling constant is unverified. This is why the map is defined on the normalized response and why we report the absolute normalization as a measurement rather than a test. (ii) The laser eigenmodes use the cold magnetized tensor rather than the warm-fluid one; the difference is below 2×10^{-3} here but would matter in a hotter or denser plasma. (iii) The reference PIC data are collisionless, two-dimensional, and short compared with the ion-acoustic damping time; the last of these forced the finite-time correction and is the largest single caveat on the absolute numbers. (iv) The orientation axis is sampled at only four values, so the near-perpendicular boundary is bracketed between 45° and 88° rather than located. (v) Cells supported by a single new run are flagged as illustrative; their seed-to-seed variance is imported from the three-seed cells rather than measured in place. (vi) The temperature axis is confounded with the distance from the reference detuning to the resonance.

Conclusions. A linear Vlasov description of magnetized cross-beam energy transfer is quantitatively predictive over a substantial and identifiable part of the parameter space, and fails in a way that its

own assumptions anticipate. Where the transfer is carried by a single ion-acoustic quasimode — co-polarized beams with the beat wavevector at 45° or less to the field — the predicted magnetization dependence matches kinetic simulation with a median relative error of 0.06 and a worst case of 0.24 from zero to 64 MG, including the splitting of the electromagnetic eigenmodes and the activation of cross-polarized transfer. That agreement, however, is mostly a test of magnetized linear optics: switching the eigenmode birefringence off in the model removes essentially all of the magnetization effect at those orientations, so the kinetic part of the theory is probed only as the beat wavevector turns perpendicular to the field. There it fails. Where magnetization fragments the mediating response into a comb of ion Bernstein resonances — near-perpendicular geometry above about 10 MG — the single-quasimode formula over-predicts transfer that the simulations show quenched, and no amount of care in the drive term repairs it. The boundary is compact: two conditions in two variables reproduce 87% of the per-cell classifications, and the two remaining swept axes, laser intensity and electron temperature, carry almost none of the boundary signal over the ranges we swept.

Two systematics cut across the whole map and are worth separating from the magnetization physics. The mediating wave in these simulations never reaches steady state, so a steady-state kernel is not directly comparable to them at all; and the detuning of peak transfer is measured at 0.58–0.72 of the linear ion-acoustic resonance in three different plasmas, an offset that we show is not caused by the in-plasma beam geometry and that we leave open. The first is an artefact of the simulation set-up and disappears in a nanosecond-scale target; the second is not, and closing it is the most useful next step — both for magnetized CBET and, since the offset is present already at $B = 0$, for CBET modelling generally.

Reproducibility. All code, all logs, the input decks and the retained field dumps of every new simulation, the frozen comparison plan with its version stamp, and the machine-readable per-cell table are included with this report; the layout is described in `proposal/README.md`. Every result quoted in the text — per-cell metrics and classifications, the compactness and mechanism statistics, the boundary-rule accuracy, and the summary counts — is generated by `proposal/code/summarize.py` into `proposal/logs/macros.tex` and read from there by the manuscript; configuration values (densities, temperatures, grid sizes, interaction times) are quoted from the input decks and the reconstruction. Each quantitative claim therefore traces to a logged value with the same number.

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A All sweep cells

Table 4 lists every cell that entered the boundary analysis. Columns: plasma (C carbon, H hydrogen; * marks a new reduced-scale run), $|B|$ in MG, θ_{kB} in degrees, T_e in keV, I in 10^{15} W/cm², initial polarization, measured channel (tot/X/O), the PIC gain and its half-spread, the $V1$ gain, the pre-registered metric ρ (capped at 9.99 for display), the classification (h = holds, b = breaks) and the attributed mechanism. Machine-readable versions of this table, of the per-configuration Vlasov output including every mechanism indicator, of the mode census and of the per-run PIC diagnostics are in `proposal/logs/`.

Table 4: All sweep cells.

pl	B	θ	T_e	I	p	ch	g_P	σ	g_V	ρ	c/mech	pl	B	θ	T_e	I	p	ch	g_P	σ	g_V	ρ	c/mech
C*	0	45	0.8	0.50	zy	tot	0.317	0.178	0.000	1.00	b P2 drive	C	50	45	0.8	0.50	zy	X	0.688	0.096	0.355	0.15	h .
C*	0	45	0.8	0.50	zy	X	0.320	0.181	0.000	1.00	b P2 drive	C	50	45	0.8	0.50	zy	O	0.200	0.074	0.293	1.41	b P1 drive
C*	0	45	0.8	0.50	zy	O	0.313	0.174	0.000	1.00	b P2 drive	C	64	45	0.8	0.50	zy	tot	0.533	0.090	0.321	0.01	h .
C*	10	45	0.8	0.25	zy	tot	-0.005	0.030	0.015	0.99	b C4 floor	C	64	45	0.8	0.50	zy	X	0.817	0.107	0.349	0.30	h .
C*	10	45	0.8	1.00	zy	tot	0.089	0.036	0.073	0.19	h .	C	64	45	0.8	0.50	zy	O	0.164	0.064	0.293	1.89	b P1 drive
C*	10	45	0.8	2.00	zy	tot	0.216	0.054	0.191	0.08	h .	C	0	45	0.8	0.50	zz	tot	1.121	0.133	0.682	0.00	h .
C*	10	45	0.8	0.50	zy	tot	0.022	0.018	0.032	0.49	b C1 steady	C	0	45	0.8	0.50	zz	X	1.122	0.133	0.682	0.00	h .
C*	10	45	0.8	0.25	zy	X	0.038	0.037	0.035	0.44	b .	C	0	45	0.8	0.50	zz	O	1.121	0.133	0.682	0.00	h .
C*	10	45	0.8	1.00	zy	X	0.226	0.038	0.162	0.08	h .	C	0	45	0.8	0.50	zz	tot	1.035	0.135	0.682	0.08	h .
C*	10	45	0.8	2.00	zy	X	0.458	0.063	0.379	0.01	h .	C	0	45	0.8	0.50	zz	X	1.035	0.138	0.666	0.06	h .
C*	10	45	0.8	0.50	zy	X	0.079	0.021	0.074	0.46	b .	C	0	45	0.8	0.50	zz	O	1.034	0.132	0.697	0.11	h .
C*	10	45	0.8	0.25	zy	O	-0.052	0.036	-0.006	0.84	b C1 steady	C	1	45	0.8	0.50	zz	tot	1.037	0.146	0.679	0.08	h .
C*	10	45	0.8	1.00	zy	O	-0.080	0.036	-0.022	0.39	b C1 steady	C	1	45	0.8	0.50	zz	X	1.039	0.151	0.647	0.02	h .
C*	10	45	0.8	2.00	zy	O	-0.126	0.044	-0.036	0.40	b C1 steady	C	1	45	0.8	0.50	zz	O	1.035	0.141	0.710	0.13	h .
C*	10	45	0.8	0.50	zy	O	-0.044	0.028	-0.011	0.45	b C1 steady	C	2	45	0.8	0.50	zz	tot	1.039	0.136	0.670	0.06	h .
C*	50	45	0.8	0.25	zy	tot	0.099	0.046	0.058	0.08	h .	C	2	45	0.8	0.50	zz	X	1.044	0.144	0.605	0.05	h .
C*	50	45	0.8	1.00	zy	tot	0.422	0.083	0.242	0.13	h .	C	2	45	0.8	0.50	zz	O	1.034	0.127	0.731	0.16	h .
C*	50	45	0.8	2.00	zy	tot	0.798	0.135	0.508	0.23	h .	C	4	45	0.8	0.50	zz	tot	1.009	0.136	0.636	0.04	h .
C*	50	45	0.8	0.50	zy	tot	0.188	0.047	0.118	0.02	h .	C	4	45	0.8	0.50	zz	X	1.016	0.151	0.507	0.18	h .
C*	50	45	0.8	0.25	zy	X	0.142	0.052	0.068	0.26	h .	C	4	45	0.8	0.50	zz	O	1.000	0.120	0.749	0.23	h .
C*	50	45	0.8	1.00	zy	X	0.600	0.103	0.274	0.31	b C1 steady	C	5	45	0.8	0.50	zz	tot	0.944	0.142	0.611	0.06	h .
C*	50	45	0.8	2.00	zy	X	1.065	0.154	0.551	0.37	b C2 depl.	C	5	45	0.8	0.50	zz	X	0.947	0.162	0.452	0.21	h .
C*	50	45	0.8	0.50	zy	X	0.284	0.059	0.136	0.25	h .	C	5	45	0.8	0.50	zz	O	0.940	0.121	0.747	0.31	b C1 steady
C*	50	45	0.8	0.25	zy	O	0.056	0.043	0.049	0.38	b C1 steady	C	8	45	0.8	0.50	zz	tot	0.864	0.160	0.519	0.01	h .
C*	50	45	0.8	1.00	zy	O	0.216	0.059	0.212	0.47	b C1 steady	C	8	45	0.8	0.50	zz	X	0.873	0.192	0.294	0.45	b C1 steady
C*	50	45	0.8	2.00	zy	O	0.456	0.103	0.465	0.24	h .	C	8	45	0.8	0.50	zz	O	0.852	0.126	0.701	0.35	b C1 steady
C*	50	45	0.8	0.50	zy	O	0.088	0.036	0.101	0.80	b C1 steady	C	10	45	0.8	0.50	zz	tot	0.722	0.149	0.453	0.03	h .
C*	0	45	0.8	0.25	zz	tot	0.192	0.050	0.124	0.00	h .	C	10	45	0.8	0.50	zz	X	0.727	0.179	0.213	0.52	b C1 steady
C*	0	45	0.8	1.00	zz	tot	0.791	0.129	0.524	0.00	h .	C	10	45	0.8	0.50	zz	O	0.715	0.118	0.645	0.48	b C1 steady
C*	0	45	0.8	2.00	zz	tot	1.367	0.193	1.124	0.00	h .	C	12	45	0.8	0.50	zz	tot	0.655	0.133	0.396	0.01	h .
C*	0	45	0.8	0.50	zz	tot	0.394	0.071	0.252	0.00	h .	C	12	45	0.8	0.50	zz	X	0.657	0.156	0.167	0.58	b C1 steady
C*	0	45	0.8	0.25	zz	X	0.192	0.051	0.124	0.00	h .	C	12	45	0.8	0.50	zz	O	0.651	0.109	0.582	0.47	b C1 steady
C*	0	45	0.8	1.00	zz	X	0.792	0.129	0.524	0.00	h .	C	14	45	0.8	0.50	zz	tot	0.525	0.123	0.356	0.11	h .
C*	0	45	0.8	2.00	zz	X	1.368	0.193	1.124	0.00	h .	C	14	45	0.8	0.50	zz	X	0.492	0.146	0.157	0.48	b C1 steady
C*	0	45	0.8	0.50	zz	X	0.395	0.071	0.252	0.00	h .	C	14	45	0.8	0.50	zz	O	0.557	0.101	0.521	0.54	b C1 steady
C*	0	45	0.8	0.25	zz	O	0.191	0.050	0.124	0.00	h .	C	16	45	0.8	0.50	zz	tot	0.449	0.078	0.333	0.22	h .
C*	0	45	0.8	1.00	zz	O	0.792	0.129	0.524	0.00	h .	C	16	45	0.8	0.50	zz	X	0.410	0.092	0.174	0.30	b C1 steady
C*	0	45	0.8	2.00	zz	O	1.368	0.193	1.124	0.00	h .	C	16	45	0.8	0.50	zz	O	0.488	0.072	0.471	0.59	b C1 steady
C*	0	45	0.8	0.50	zz	O	0.395	0.071	0.252	0.00	h .	C	16	45	0.8	0.50	zz	tot	0.437	0.077	0.330	0.24	h .
C*	10	45	0.8	0.25	zz	tot	0.212	0.043	0.109	0.20	h .	C	16	45	0.8	0.50	zz	X	0.407	0.079	0.180	0.27	h .
C*	10	45	0.8	1.00	zz	tot	0.666	0.109	0.467	0.06	h .	C	16	45	0.8	0.50	zz	O	0.468	0.075	0.461	0.62	b C1 steady
C*	10	45	0.8	2.00	zz	tot	1.245	0.178	1.014	0.01	h .	C	20	45	0.8	0.50	zz	tot	0.441	0.076	0.331	0.23	h .
C*	10	45	0.8	0.50	zz	tot	0.344	0.070	0.223	0.01	h .	C	20	45	0.8	0.50	zz	X	0.448	0.082	0.232	0.15	h .
C*	10	45	0.8	0.25	zz	X	0.202	0.052	0.089	0.31	b C1 steady	C	20	45	0.8	0.50	zz	O	0.435	0.071	0.422	0.59	b C1 steady
C*	10	45	0.8	1.00	zz	X	0.615	0.119	0.384	0.06	h .	C	24	45	0.8	0.50	zz	tot	0.457	0.079	0.343	0.23	h .
C*	10	45	0.8	2.00	zz	X	1.176	0.189	0.848	0.12	h .	C	24	45	0.8	0.50	zz	X	0.485	0.087	0.264	0.11	h .
C*	10	45	0.8	0.50	zz	X	0.320	0.077	0.183	0.11	h .	C	24	45	0.8	0.50	zz	O	0.427	0.071	0.418	0.61	b C1 steady
C*	10	45	0.8	0.25	zz	O	0.222	0.037	0.129	0.10	h .	C	28	45	0.8	0.50	zz	tot	0.468	0.080	0.340	0.19	h .
C*	10	45	0.8	1.00	zz	O	0.713	0.100	0.544	0.15	h .	C	28	45	0.8	0.50	zz	X	0.535	0.084	0.262	0.20	h .
C*	10	45	0.8	2.00	zz	O	1.307	0.167	1.158	0.08	h .	C	28	45	0.8	0.50	zz	O	0.393	0.082	0.414	0.73	b C1 steady
C*	10	45	0.8	0.50	zz	O	0.367	0.063	0.263	0.12	h .	C	32	45	0.8	0.50	zz	tot	0.449	0.092	0.328	0.20	h .
C*	50	45	0.8	0.25	zz	tot	0.102	0.033	0.064	0.03	h .	C	32	45	0.8	0.50	zz	X	0.529	0.099	0.258	0.20	h .
C*	50	45	0.8	1.00	zz	tot	0.447	0.085	0.266	0.10	h .	C	32	45	0.8	0.50	zz	O	0.359	0.083	0.394	0.80	b C1 steady
C*	50	45	0.8	2.00	zz	tot	0.849	0.129	0.559	0.20	h .	C	50	45	0.8	0.50	zz	tot	0.493	0.100	0.323	0.08	h .
C*	50	45	0.8	0.50	zz	tot	0.208	0.047	0.130	0.03	h .	C	50	45	0.8	0.50	zz	X	0.664	0.109	0.287	0.29	h .
C*	50	45	0.8	0.25	zz	X	0.116	0.038	0.057	0.24	h .	C	50	45	0.8	0.50	zz	O	0.278	0.087	0.359	1.12	b P1 drive
C*	50	45	0.8	1.00	zz	X	0.523	0.106	0.231	0.33	b C1 steady	C	64	45	0.8	0.50	zz	tot	0.558	0.089	0.322	0.05	h .
C*	50	45	0.8	2.00	zz	X	0.988	0.149	0.470	0.42	b C1 steady	C	64	45	0.8								

pl	B	θ	T_e	I	p	ch	g_P	σ	g_V	ρ	c/mech	pl	B	θ	T_e	I	p	ch	g_P	σ	g_V	ρ	c/mech
C* 10	90	0.8	0.50	zz	O		0.272	0.041	-0.244	2.40	b C3 L_x	C 4	88	0.8	0.50	zy	tot		-0.118	0.073	0.070	1.39	b -
C* 50	90	0.8	0.50	zz	tot		0.050	0.018	0.003	0.77	b C1 steady	C 4	88	0.8	0.50	zy	X		0.064	0.083	0.203	1.60	b C4 floor
C* 50	90	0.8	0.50	zz	X		0.035	0.029	0.003	0.53	b C1 steady	C 4	88	0.8	0.50	zy	O		-0.342	0.130	-0.085	0.59	b C1 steady
C* 50	90	0.8	0.50	zz	O		0.065	0.021	0.003	0.93	b C1 steady	C 5	88	0.8	0.50	zy	tot		-0.083	0.071	0.097	1.44	b -
C 0	0	0.8	0.50	zy	tot		-0.190	0.385	0.000	1.00	b C4 floor	C 5	88	0.8	0.50	zy	X		0.192	0.078	0.236	1.02	b C1 steady
C 0	0	0.8	0.50	zy	X		-0.201	0.311	0.000	1.00	b C4 floor	C 5	88	0.8	0.50	zy	O		-0.460	0.062	-0.065	0.77	b C1 steady
C 0	0	0.8	0.50	zy	O		-0.186	0.459	0.000	1.00	b C4 floor	C 8	88	0.8	0.50	zy	tot		-0.138	0.107	0.175	2.53	b -
C 0	0	0.8	0.50	zy	tot		-0.240	0.118	0.000	1.00	b -	C 8	88	0.8	0.50	zy	X		0.041	0.131	0.292	2.61	b C4 floor
C 0	0	0.8	0.50	zy	X		-0.280	0.197	0.007	1.04	b -	C 8	88	0.8	0.50	zy	O		-0.352	0.071	0.041	1.19	b C1 steady
C 0	0	0.8	0.50	zy	O		-0.205	0.075	-0.007	0.95	b -	C 10	88	0.8	0.50	zy	tot		-0.218	0.112	0.214	2.61	b -
C 1	0	0.8	0.50	zy	tot		-0.389	0.221	0.000	1.00	b -	C 10	88	0.8	0.50	zy	X		-0.159	0.109	0.308	3.95	b -
C 1	0	0.8	0.50	zy	X		-0.414	0.263	0.014	1.06	b C1 steady	C 10	88	0.8	0.50	zy	O		-0.279	0.115	0.111	1.66	b C1 steady
C 1	0	0.8	0.50	zy	O		-0.367	0.189	-0.013	0.94	b C1 steady	C 16	88	0.8	0.50	zy	tot		-0.147	0.100	0.248	3.30	b P4 modes
C 2	0	0.8	0.50	zy	tot		-0.394	0.218	0.002	1.01	b -	C 16	88	0.8	0.50	zy	X		-0.219	0.103	0.282	3.12	b P4 modes
C 2	0	0.8	0.50	zy	X		-0.380	0.191	0.029	1.13	b C1 steady	C 16	88	0.8	0.50	zy	O		-0.080	0.097	0.214	2.56	b C4 floor
C 2	0	0.8	0.50	zy	O		-0.410	0.249	-0.026	0.90	b C1 steady	C 32	88	0.8	0.50	zy	tot		-0.117	0.095	0.238	3.02	b P4 modes
C 4	0	0.8	0.50	zy	tot		-0.234	0.111	0.007	1.05	b -	C 32	88	0.8	0.50	zy	X		-0.184	0.087	0.254	3.27	b P4 modes
C 4	0	0.8	0.50	zy	X		-0.317	0.120	0.061	1.31	b C1 steady	C 32	88	0.8	0.50	zy	O		-0.054	0.102	0.222	2.49	b C4 floor
C 4	0	0.8	0.50	zy	O		-0.161	0.104	-0.048	0.49	b C1 steady	C 50	88	0.8	0.50	zy	tot		-0.097	0.129	0.234	2.86	b C4 floor
C 5	0	0.8	0.50	zy	tot		-0.240	0.114	0.010	1.07	b -	C 50	88	0.8	0.50	zy	X		-0.146	0.166	0.244	3.25	b C4 floor
C 5	0	0.8	0.50	zy	X		-0.250	0.130	0.077	1.51	b C1 steady	C 50	88	0.8	0.50	zy	O		-0.052	0.096	0.224	2.50	b C4 floor
C 5	0	0.8	0.50	zy	O		-0.231	0.099	-0.057	0.59	b C1 steady	C 64	88	0.8	0.50	zy	tot		-0.071	0.130	0.233	2.71	b C4 floor
C 8	0	0.8	0.50	zy	tot		-0.190	0.114	0.026	1.22	b -	C 64	88	0.8	0.50	zy	X		-0.140	0.185	0.242	3.20	b C4 floor
C 8	0	0.8	0.50	zy	X		-0.188	0.182	0.130	2.13	b C1 steady	C 64	88	0.8	0.50	zy	O		-0.007	0.082	0.225	2.24	b C4 floor
C 8	0	0.8	0.50	zy	O		-0.196	0.074	-0.080	0.33	b C1 steady	C 80	88	0.8	0.50	zy	tot		-0.068	0.092	0.231	2.66	b C4 floor
C 10	0	0.8	0.50	zy	tot		-0.136	0.100	0.040	1.20	b -	C 80	88	0.8	0.50	zy	X		-0.128	0.135	0.238	3.08	b C4 floor
C 10	0	0.8	0.50	zy	X		-0.052	0.161	0.166	1.93	b C4 floor	C 80	88	0.8	0.50	zy	O		-0.014	0.053	0.223	2.26	b C4 floor
C 10	0	0.8	0.50	zy	O		-0.222	0.084	-0.089	0.34	b C1 steady	C 0	88	0.8	0.50	zz	tot		1.121	0.133	0.682	0.00	h -
C 16	0	0.8	0.50	zy	tot		-0.016	0.089	0.093	1.00	b C4 floor	C 0	88	0.8	0.50	zz	X		1.122	0.133	0.682	0.00	h -
C 16	0	0.8	0.50	zy	X		0.111	0.087	0.275	2.03	b C1 steady	C 0	88	0.8	0.50	zz	O		1.121	0.133	0.682	0.00	h -
C 16	0	0.8	0.50	zy	O		-0.134	0.098	-0.089	0.07	h -	C 0	88	0.8	0.50	zz	tot		1.046	0.155	0.681	0.07	h -
C 32	0	0.8	0.50	zy	tot		0.343	0.079	0.258	0.24	h -	C 0	88	0.8	0.50	zz	X		1.047	0.158	0.663	0.04	h -
C 32	0	0.8	0.50	zy	X		0.573	0.087	0.504	0.44	b C1 steady	C 0	88	0.8	0.50	zz	O		1.045	0.152	0.698	0.10	h -
C 32	0	0.8	0.50	zy	O		0.151	0.072	0.047	0.44	b C1 steady	C 1	88	0.8	0.50	zz	tot		1.088	0.188	0.666	0.01	h -
C 50	0	0.8	0.50	zy	tot		0.501	0.077	0.337	0.10	h -	C 1	88	0.8	0.50	zz	X		1.093	0.194	0.630	0.05	h -
C 50	0	0.8	0.50	zy	X		0.869	0.094	0.563	0.06	h -	C 1	88	0.8	0.50	zz	O		1.084	0.182	0.701	0.06	h -
C 50	0	0.8	0.50	zy	O		0.220	0.073	0.187	0.39	b P1 drive	C 2	88	0.8	0.50	zz	tot		1.030	0.131	0.629	0.00	h -
C 64	0	0.8	0.50	zy	tot		0.570	0.119	0.307	0.12	h -	C 2	88	0.8	0.50	zz	X		1.040	0.141	0.558	0.12	h -
C 64	0	0.8	0.50	zy	X		1.038	0.107	0.522	0.17	h -	C 2	88	0.8	0.50	zz	O		1.020	0.120	0.695	0.12	h -
C 64	0	0.8	0.50	zy	O		0.247	0.132	0.188	0.25	h -	C 4	88	0.8	0.50	zz	tot		0.819	0.101	0.540	0.08	h -
C 0	0	0.8	0.50	zz	tot		1.121	0.133	0.682	0.00	h -	C 4	88	0.8	0.50	zz	X		0.824	0.120	0.413	0.18	h -
C 0	0	0.8	0.50	zz	X		1.122	0.133	0.682	0.00	h -	C 4	88	0.8	0.50	zz	O		0.812	0.080	0.652	0.32	b -
C 0	0	0.8	0.50	zz	O		1.121	0.133	0.682	0.00	h -	C 5	88	0.8	0.50	zz	tot		0.732	0.091	0.492	0.10	h -
C 0	0	0.8	0.50	zz	tot		1.041	0.163	0.682	0.08	h -	C 5	88	0.8	0.50	zz	X		0.729	0.108	0.347	0.22	h -
C 0	0	0.8	0.50	zz	X		1.042	0.165	0.677	0.07	h -	C 5	88	0.8	0.50	zz	O		0.733	0.074	0.617	0.38	b -
C 0	0	0.8	0.50	zz	O		1.041	0.162	0.687	0.08	h -	C 8	88	0.8	0.50	zz	tot		0.321	0.085	0.367	0.88	b -
C 1	0	0.8	0.50	zz	tot		1.050	0.106	0.682	0.07	h -	C 8	88	0.8	0.50	zz	X		0.277	0.070	0.217	0.29	h -
C 1	0	0.8	0.50	zz	X		1.050	0.106	0.673	0.05	h -	C 8	88	0.8	0.50	zz	O		0.362	0.107	0.497	1.25	b -
C 1	0	0.8	0.50	zz	O		1.049	0.105	0.692	0.08	h -	C 10	88	0.8	0.50	zz	tot		0.125	0.149	0.307	2.26	b C4 floor
C 2	0	0.8	0.50	zz	tot		1.073	0.130	0.681	0.04	h -	C 10	88	0.8	0.50	zz	X		0.095	0.154	0.175	1.15	b C4 floor
C 2	0	0.8	0.50	zz	X		1.075	0.132	0.662	0.01	h -	C 10	88	0.8	0.50	zz	O		0.154	0.146	0.424	3.23	b -
C 2	0	0.8	0.50	zz	O		1.072	0.127	0.700	0.07	h -	C 16	88	0.8	0.50	zz	tot		0.032	0.103	0.239	2.15	b C4 floor
C 4	0	0.8	0.50	zz	tot		1.028	0.156	0.678	0.08	h -	C 16	88	0.8	0.50	zz	X		0.032	0.079	0.180	1.57	b C4 floor
C 4	0	0.8	0.50	zz	X		1.029	0.161	0.640	0.02	h -	C 16	88	0.8	0.50	zz	O		0.032	0.127	0.295	2.69	b C4 floor
C 4	0	0.8	0.50	zz	O		1.028	0.152	0.716	0.14	h -	C 32	88	0.8	0.50	zz	tot		-0.046	0.114	0.236	2.58	b C4 floor
C 5	0	0.8	0.50	zz	tot		1.030	0.133	0.676	0.08	h -	C 32	88	0.8	0.50	zz	X		-0.044	0.108	0.211	2.32	b C4 floor
C 5	0	0.8	0.50	zz	X		1.032	0.139	0.629	0.00	h -	C 32	88	0.8	0.50	zz	O		-0.048	0.120	0.261	2.83	b C4 floor
C 5	0	0.8	0.50	zz	O		1.029	0.127	0.723	0.15	h -	C 50	88	0.8	0.50	zz	tot		-0.039	0.131	0.236	2.54	b C4 floor
C 8	0	0.8	0.50	zz	tot		1.050	0.156	0.665	0.04	h -	C 50	88	0.8	0.50	zz	X		-0.040	0.138	0.221	2.40	b C4 floor
C 8	0	0.8	0.50	zz	X		1.054	0.165	0.592	0.08	h -	C 50	88	0.8	0.50	zz	O		-0.039	0.124	0.250	2.67	b C4 floor
C 8	0	0.8	0.50	zz	O		1.046	0.145	0.740	0.16	h -	C 64	88	0.8	0.50	zz	tot		-0.058	0.119	0.234	2.63	b C4 floor
C 10	0	0.8	0.50	zz	tot		1.026	0.143	0.656	0.05	h -	C 64	88	0.8	0.50	zz	X		-0.068	0.167	0.224	2.59	b C4 floor
C 10	0	0.8	0.50	zz	X		1.032	0.153	0.566	0.10	h -	C 64	88	0.8	0.50	zz	O		-0.047	0.071	0.245	2.67	b C4 floor
C 10	0	0.8	0.50	zz	O		1.019	0.130	0.748	0.21	h -	H* 0	71	0.8	1.00	zy	tot		-0.009	0.035			

pl	B	θ	T_e	I	p	ch	g_P	σ	g_V	ρ	c/mech	pl	B	θ	T_e	I	p	ch	g_P	σ	g_V	ρ	c/mech
C	4	45	0.8	0.50	zy	O	-0.112	0.070	-0.100	0.32	b C1 steady	H	22	71	0.4	1.00	zy	tot	0.697	0.000	0.956	0.46	b P3 ions
C	5	45	0.8	0.50	zy	tot	-0.024	0.077	0.100	1.12	b C4 floor	H	25	71	0.4	1.00	zy	tot	0.663	0.000	0.938	0.50	b P3 ions
C	5	45	0.8	0.50	zy	X	0.010	0.078	0.260	2.48	b C4 floor	H	28	71	0.4	1.00	zy	tot	0.638	0.000	0.904	0.50	b P3 ions
C	5	45	0.8	0.50	zy	O	-0.062	0.095	-0.093	0.54	b C4 floor	H	30	71	0.4	1.00	zy	tot	0.723	0.000	0.864	0.27	h .
C	8	45	0.8	0.50	zy	tot	0.158	0.076	0.204	1.05	b .	H	0	71	0.4	1.00	zz	tot	2.047	0.000	1.927	0.00	h .
C	8	45	0.8	0.50	zy	X	0.186	0.088	0.377	2.33	b C1 steady	H	2	71	0.4	1.00	zz	tot	1.969	0.000	1.902	0.03	h .
C	8	45	0.8	0.50	zy	O	0.127	0.080	-0.008	0.83	b C1 steady	H	5	71	0.4	1.00	zz	tot	1.781	0.000	1.806	0.08	h .
C	10	45	0.8	0.50	zy	tot	0.320	0.078	0.264	0.35	b .	H	8	71	0.4	1.00	zz	tot	1.666	0.000	1.700	0.08	h .
C	10	45	0.8	0.50	zy	X	0.425	0.085	0.422	0.63	b C1 steady	H	10	71	0.4	1.00	zz	tot	1.286	0.000	1.576	0.30	b C2 depl.
C	10	45	0.8	0.50	zy	O	0.203	0.082	0.075	0.40	b C1 steady	H	12	71	0.4	1.00	zz	tot	1.367	0.000	1.434	0.11	h .
C	12	45	0.8	0.50	zy	tot	0.365	0.092	0.308	0.38	b .	H	15	71	0.4	1.00	zz	tot	1.036	0.000	1.282	0.32	b C2 depl.
C	12	45	0.8	0.50	zy	X	0.493	0.094	0.441	0.47	b C1 steady	H	18	71	0.4	1.00	zz	tot	0.971	0.000	1.135	0.24	h .
C	12	45	0.8	0.50	zy	O	0.220	0.089	0.154	0.15	h .	H	20	71	0.4	1.00	zz	tot	0.969	0.000	1.008	0.11	h .
C	14	45	0.8	0.50	zy	tot	0.418	0.089	0.334	0.31	b .	H	22	71	0.4	1.00	zz	tot	0.783	0.000	0.917	0.24	h .
C	14	45	0.8	0.50	zy	X	0.507	0.118	0.438	0.42	b C1 steady	H	25	71	0.4	1.00	zz	tot	0.761	0.000	0.869	0.21	h .
C	14	45	0.8	0.50	zy	O	0.320	0.073	0.217	0.11	h .	H	28	71	0.4	1.00	zz	tot	0.616	0.000	0.857	0.48	b P3 ions
C	16	45	0.8	0.50	zy	tot	0.442	0.099	0.343	0.28	h .	H	30	71	0.4	1.00	zz	tot	0.717	0.000	0.868	0.28	h .
C	16	45	0.8	0.50	zy	X	0.556	0.132	0.421	0.24	h .	H	0	71	1.2	1.00	zy	tot	-0.183	0.015	0.000	1.00	b P2 drive
C	16	45	0.8	0.50	zy	O	0.313	0.073	0.259	0.36	b .	H	2	71	1.2	1.00	zy	tot	-0.157	0.013	0.002	1.06	b .
C	16	45	0.8	0.50	zy	tot	0.441	0.078	0.343	0.28	h .	H	5	71	1.2	1.00	zy	tot	-0.079	0.023	0.006	1.19	b .
C	16	45	0.8	0.50	zy	X	0.564	0.096	0.415	0.21	h .	H	8	71	1.2	1.00	zy	tot	-0.091	0.064	0.014	1.81	b .
C	16	45	0.8	0.50	zy	O	0.302	0.073	0.266	0.45	b .	H	10	71	1.2	1.00	zy	tot	0.067	0.020	0.024	0.88	b .
C	20	45	0.8	0.50	zy	tot	0.444	0.085	0.330	0.22	h .	H	12	71	1.2	1.00	zy	tot	0.090	0.049	0.035	1.33	b .
C	20	45	0.8	0.50	zy	X	0.586	0.116	0.375	0.05	h .	H	18	71	1.2	1.00	zy	tot	0.218	0.062	0.054	0.57	b .
C	20	45	0.8	0.50	zy	O	0.280	0.077	0.285	0.67	b .	H	20	71	1.2	1.00	zy	tot	0.218	0.014	0.060	0.74	b .
C	24	45	0.8	0.50	zy	tot	0.415	0.078	0.314	0.24	h .	H	0	71	1.2	1.00	zz	tot	0.668	0.014	0.105	0.00	h .
C	24	45	0.8	0.50	zy	X	0.549	0.107	0.355	0.06	h .	H	2	71	1.2	1.00	zz	tot	0.609	0.097	0.104	0.08	h .
C	24	45	0.8	0.50	zy	O	0.262	0.094	0.271	0.70	b .	H	5	71	1.2	1.00	zz	tot	0.754	0.045	0.103	0.13	h .
C	28	45	0.8	0.50	zy	tot	0.435	0.080	0.315	0.19	h .	H	8	71	1.2	1.00	zz	tot	0.592	0.084	0.103	0.10	h .
C	28	45	0.8	0.50	zy	X	0.599	0.093	0.363	0.00	h .	H	10	71	1.2	1.00	zz	tot	0.516	0.016	0.098	0.21	h .
C	28	45	0.8	0.50	zy	O	0.244	0.070	0.267	0.80	b .	H	12	71	1.2	1.00	zz	tot	0.307	0.014	0.091	0.88	b .
C	32	45	0.8	0.50	zy	tot	0.444	0.080	0.325	0.20	h .	H	15	71	1.2	1.00	zz	tot	0.540	0.002	0.083	0.02	h .
C	32	45	0.8	0.50	zy	X	0.596	0.088	0.369	0.02	h .	H	18	71	1.2	1.00	zz	tot	0.287	0.003	0.076	0.68	b .
C	32	45	0.8	0.50	zy	O	0.270	0.069	0.281	0.71	b P1 drive	H	20	71	1.2	1.00	zz	tot	0.278	0.014	0.070	0.61	b .
C	50	45	0.8	0.50	zy	tot	0.468	0.081	0.324	0.14	h .												